

1/✓ Calculate the answer in Engineering units: $(1.7 \times 10^3) \times (2.2 \times 10^{-5})$

$$(1.7 \times 10^3) \times (2.2 \times 10^{-5})$$

$$3.74 \times 10^{-2}$$

$$37.4 \times 10^{-3}$$

$$37.4 \text{ E } -3$$

1/✓ If the power consumed in a resistor is 20 W and the current through it is 2 Amps, find the value of the resistor in ohms.

$$P = I^2 R$$

$$20 = (2)^2 R$$

$$R = \frac{20}{4}$$

$$R = 5 \Omega \checkmark$$

1/✓ Gallium has Atomic Number 31. Calculate the number of electrons in its outermost orbitals s and p.

$$1s^2 \quad 2s^2 \quad 2p^6 \quad 3s^2 \quad 3p^6 \quad 3d^{10} \quad 4s^2 \quad 4p^1$$

k

k

m

n

$$1s^2 \quad 2s^2 \quad 2p^6 \quad 3s^2 \quad 3p^6 \quad 3d^{10} \quad 4s^2 \quad 4p^1$$

2 electrons in 4s subshell & 1 electron in 4p subshell.

1/✓ Draw diagram of Crystalline structures in solids, showing the atoms.



1/✓ Why is Valence Shell more important than inner shells of an atom.

Because only the valence shell is involved in the conduction, overlapping and merging of energy levels to form bands only occur in valence shell & only valence shell is involved in bonding.



If $J = \sigma E$, where σ has units $1/\text{ohm.cm}$ and E has units volt/cm .

Work out the units of J .

$$J = \sigma E$$

$$= \frac{1}{\text{ohm.cm}} \times \frac{\text{volt}}{\text{cm}}$$

$$= \frac{\text{volt}}{\text{ohm.cm}^2}$$

$$J = \frac{\text{ohm}}{\text{Amp}} \times \frac{1}{\text{cm}^2}$$

since $V = IR$
 $I = \frac{V}{R}$
 $A = \frac{\text{Volt}}{\text{ohm}}$

$\sigma = n \cdot q \cdot \mu$ If units of n are $1/\text{cm}^3$, q are coulomb and μ are $\text{cm}^2/(\text{Volt} \cdot \text{sec})$

Prove that units of σ are $1/(\text{ohm} \cdot \text{cm})$

since $J = \frac{Q}{t} = \frac{C}{s}$
 $A = \frac{C}{s}$

$$\sigma = n \cdot q \cdot \mu$$

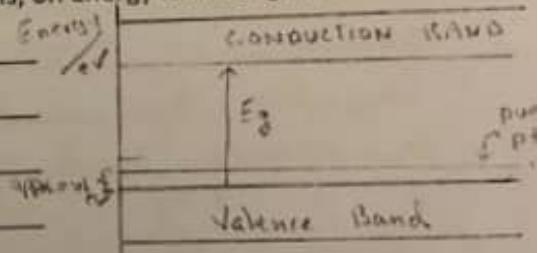
$$\frac{1}{\text{ohm.cm}} = \frac{1}{\text{cm}^3} \times \frac{C}{V} \times \frac{\text{cm}^2}{V \cdot s}$$

since $V = IR$
 $\frac{1}{R} = \frac{I}{V}$ thus $\frac{1}{\Omega} = \frac{A}{V}$

$$\frac{1}{\text{ohm.cm}} = \frac{1}{\text{cm}^3} \times \frac{A}{V} \times \frac{\text{cm}^2}{V \cdot s} = \frac{1}{\text{ohm.cm}}$$

hence proven.

Show the energy level of hole created by P-type impurity atoms, on Energy level Diagram.



Describe what happens during the Recombination process?

Recombination process is the opposite of the ionization process. Due to decreasing temperature, e^- in conduction band loose energy E fall back into the valence band recombining with p (holes) & thus annihilation of e^- & p .

Use Law of Mass Action to calculate concentration of minority carriers in N-type semiconductor.

law of Mass Action

$$n \times p = (n_i)^2$$

$$n \approx N_D \quad \text{in N-type}$$

$$p = \frac{(n_i)^2}{N_D}$$



Why does Silicon (Atomic No = 14) form Covalent bond with Phosphorus (Atomic No = 15)?

Si_{14} forms 4 covalent bond with P_{15} to achieve stable structure and P_{15} then donates 1 free electron to the structure. P_{15} is the P donor impurity to form an N-type extrinsic semiconductor.

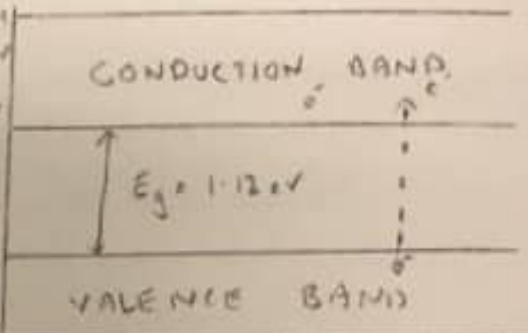
Gravitational force is considered a Universal Force.

Give names of two other Universal forces.

- Magnetic force of attraction & repulsion
- Electric force of attraction & repulsion.

Conductivity of intrinsic Silicon increases with temperature. How is this explained by Band Theory of Solids?

e^- in valence band gain energy, overcome the forbidden energy gap E_g & excite to the conduction band where they are free to move. The higher the temperature the greater number of e^- excite to the conduction band. μ of e^- also increases.



Give the Energy Gap for Silicon and Copper.

$$\text{Si} = 1.12 \text{ eV}$$

$$\text{Cu} \approx 0.00 \text{ eV} \text{ almost negligible.}$$

If the resistivity of Silicon is $2 \times 10^5 \text{ ohm-cm}$. Find its Conductivity

$$R = \frac{1}{G}$$

$$G = \frac{1}{R}$$

$$\frac{\sigma A}{AL} = \frac{1}{\frac{\rho L}{A}}$$

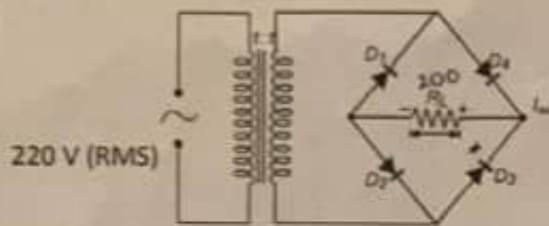
$$\frac{\sigma A}{L} = \frac{A}{\rho L}$$

$$\sigma = \frac{1}{\rho}$$

$$\sigma = \frac{1}{2 \times 10^5} = 0.5 \times 10^{-5} \text{ } \checkmark$$

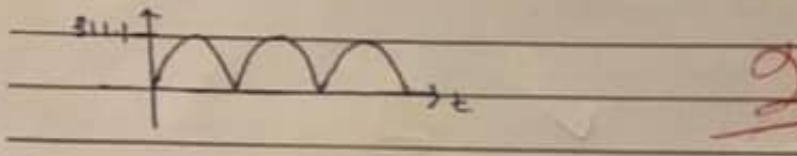


The circuit of bridge rectifier is given below. The input voltage is 220 Volt RMS Frequency 50 Hz. Load resistor $R_L = 200 \text{ ohm}$. Answer the questions that follow.



Draw voltage waveform across resistor R_L showing maximum value.

$$V_{rms} = V_m / \sqrt{2} \rightarrow V_m = 311.1 \text{ volts}$$



Find the DC voltage across R_L .

$$V_{dc} = 2V_m / \pi$$

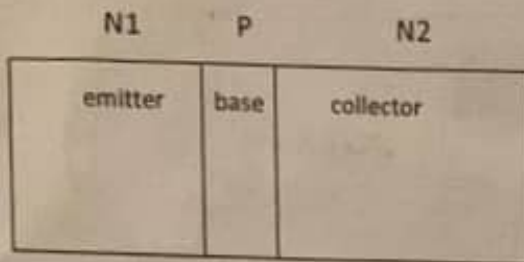
$$= 2(311.1) / \pi = 198.0 \text{ volts}$$



Find the power dissipated in R_L .

$$\text{Power} = \frac{(198.0)^2}{200} = 196.0 \text{ watt}$$

Consider the construction diagram of a NPN transistor. Answer questions that follow.



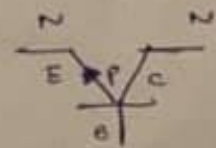
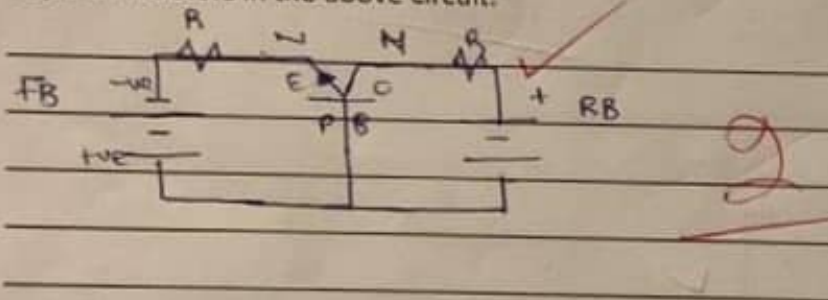
Give the typical concentration of majority carriers in N1, P and N2 regions

$$N1 : 10^{17} / \text{cm}^3$$

$$P : 10^5 / \text{cm}^3$$

$$N2 : 10^{13} / \text{cm}^3$$

Using circuit symbol of transistor, connect emitter base junction in forward bias and collector-base junction in reverse bias. Also connect required resistors in the above circuit.



If $I_E = 8 \text{ mA}$, $I_C = 7 \text{ mA}$. Find the value of current gains α and β

$$\alpha = I_C / I_E$$

$$= 7 \times 10^{-3} / 8 \times 10^{-3}$$

$$= 0.875$$

$$\beta = I_C / I_B$$

$$= 7 \times 10^{-3} / 1 \times 10^{-3}$$

$$= 7$$

$$I_E = I_B + I_C$$

$$I_B = I_E - I_C$$

$$= (8 - 7) \text{ mA}$$

$$= 1 \times 10^{-3}$$

Applied Physics :-

Notations for physical Quantities :-

Decimal

45.37

Scientific

4.537×10^1

(multiple of any
number)

0.4537×10^2

Engineering

(power of multiple of 3)

0.04537×10^3

$\Rightarrow 10^{15}, 10^{12}, 10^9, 10^6, 10^3$

10^{15} peta (P)

10^{-3} milli (m)

10^{12} tera (T)

10^{-6} micro (μ)

10^9 Giga (G)

10^{-9} nano (n)

10^6 Mega (M)

10^{-12} pico (p)

10^3 kilo (K)

10^{-15} femto (f)

$$\Rightarrow 23000 \times 0.000003$$

$$23 \times 10^3 \times 3 \times 10^{-6}$$

$$[23E3] \times 3E-6$$

mantissa
exponent

$\Rightarrow$ Speed of a super computer is measured in peta (10^{15})

$\Rightarrow$ RAM memory of super computer and hard disk storage of server is measured in Tera (10^{12})

$\Rightarrow$ Memory in mobile and mobile data/downloading speed is in Giga (10^9) e.g. 4G, 5G = 5×10^9 per seconds.

$\Rightarrow$ Clock speed of CPU is measured in Mega Hertz.

Analog frequency
Hertz (cycles/second)

Digital frequency
Bits/second

e.g. FM radio 88 Mega Hertz

TV signals 54 Mega Hertz - 210 Mega Hertz

Mega Bytes, Memory of a computer or RAM is in terms of Mega Bytes

Normal processor $\Rightarrow$ MIPS (million instructions per second)

Speed of pentium processor = 188 MIPS = 188×10^6 instructions/second

Core i7 = 82,300 MIPS

Kilo 10^3 ; High voltage transmission

e.g. 11kV $\Rightarrow$ voltage on street transformer

220V $\Rightarrow$ house

$\Rightarrow$ Medium wave radio's frequency $\Rightarrow$ 1630 kHz

$\Rightarrow$ Normal household voltage $\Rightarrow$ 5 Ampere

10^{-3} milli ampere (current)

charging current of mobile charger

Forward current of diode is in volt. (milli volts [semiconductor devices])

^{micro}
 10^{-6} μ voltage in CMOS devices

Reverse current of p-n junctions

Rating of capacitor μ Farad

10^{-9} nano ; current in CMOS devices (The machines for space exploration carry CMOS devices)

CMOS $\Rightarrow$ complimentary metal oxide semi conductor.

Analogy of Dam :-

Height of Dam $\longleftrightarrow$ voltage

~~Flow of water~~ speed of water $\longleftrightarrow$ current

Total water flowing out $\longleftrightarrow$ charge

Opening of split ways $\longleftrightarrow$ Resistance

* Power = 100 Megawatt = 100×10^6 watt

suppose voltage of AC = 220 volt

$$I = \frac{P}{V} = \frac{100 \times 10^6}{220 \approx 200} = 0.5 \times 10^6 = 500 \text{ KAmperes}$$

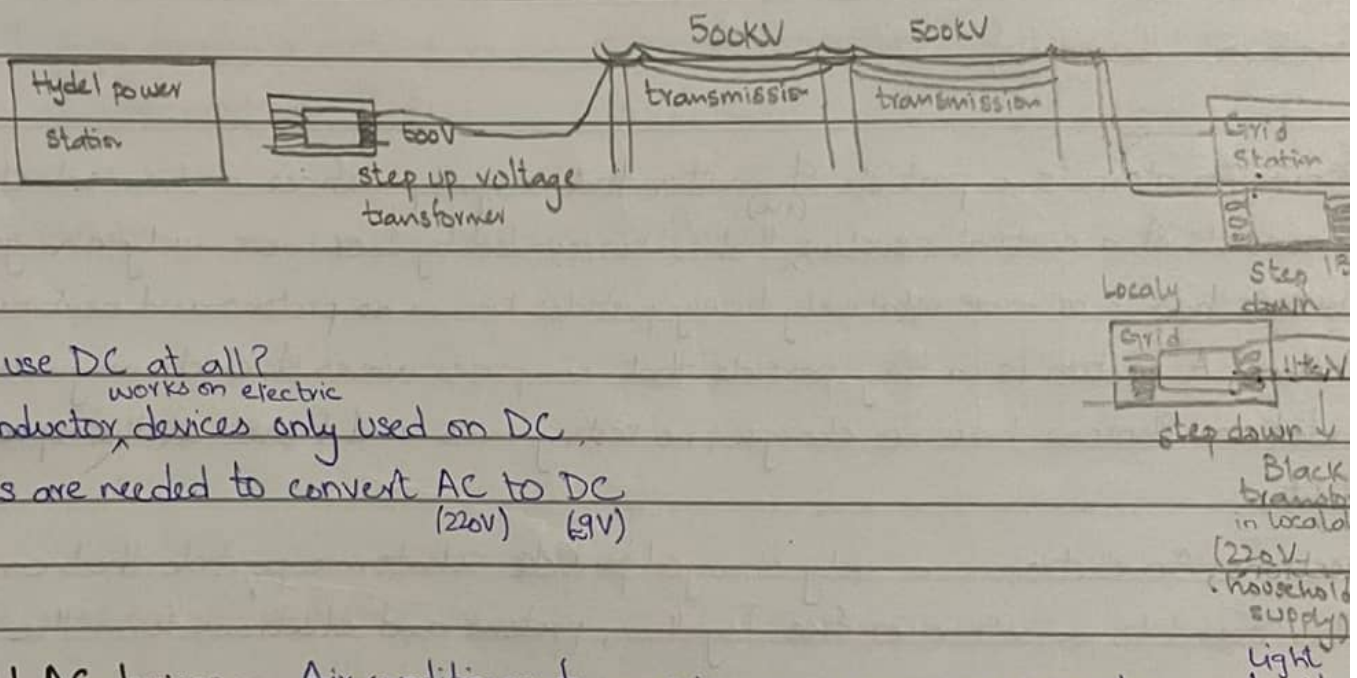
(in order to carry this current you require thick cables silver and platinum // (DC))

⇒ Use step up transformers ⇒ 220V to 500K volts

$$I = \frac{P}{V} = \frac{100 \times 10^6}{500 \times 10^3} = 0.2 \times 10^3 = 200 \text{ Amps (AC)}$$

(manageable size of cables used)

Line loss
⇒ Resistance of current ⇒ $I^2 R$ hence AC has less loss of current



→ Why use DC at all?

works on electric

Semiconductor devices only used on DC

Circuits are needed to convert AC to DC
(220V) (9V)

Household AC devices :- Air conditioner, fan, microwave oven, vacuum cleaner, bulb

TV (AC to DC converter)

Household DC devices :- Mobile phone, wall clock, LED bulb

Free electrons; The electrons which are not attached to the nucleus of an atom and free to move when external energy ~~to~~ is applied are called free electrons.

Valence Band; It is the band of electron orbitals that electrons can jump out of moving into the conduction band when excited. The valence band is simply the outermost electron orbital of an atom of any specific material that e^- actually occupies.

Conduction Band; It is the band of electron orbitals that electrons can jump up into from the valence band ~~and~~ when excited. When the e^- are here, they have enough energy to move freely in the material.

Ionization; It is a measure of the capability of an element to enter into chemical reactions requiring ion formation or donation of electrons.

Orbital; Region around a nucleus in an atom or molecule that may contain zero, one, or two electrons.

~~Insulator~~ **Insulator;** do not allow current to flow

Intrinsic semiconductor;

Extrinsic ~~insulator~~ semiconductor;

Conductor; Allows current to flow

Silicon;

*moving like planets

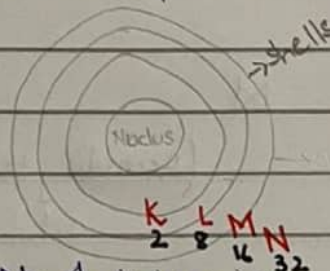
Group III, IV, V, VI

3) Semi conductor ; partially allows current to flow, e.g. germanium, silicon, phosphorus

⇒ All three are determined by atomic structure of the elements

Atomic Structure :-

Dalton's Atom → Rutherford Atom → Bohr's Atom → Planck's Atom / Einstein's Atom

Atoms are individual/smallest particle (1805)	→ Nuclear (Protons, neutrons) → Planetary electrons * → Subatomic particles  → No. of electrons in a shell is $2n^2$; $K = 2 \times 1^2 = 2$ $L = 2 \times 2^2 = 8$ $M = 2 \times 3^2 = 18$ $N = 2 \times 4^2 = 32$ → No. of electrons is equal to no. of protons (ve) → Atom is electrically neutral → Stable structure (2 outer electrons in shell along with 2 protons they don't interact with others) e.g. neon	→ discovery of subshell or orbitals → A shell is divided into subshells (s, p, d, f) 2, 6, 10, 14 Example of neon; Atomic no. = 10 Rutherford = $K=2$ $L=8$ Bohr's atom = $S=2$ $K=2$ $\begin{cases} P=0 \\ d=0 \\ f=0 \end{cases}$ $\begin{cases} s=2 \\ p=6 \\ d=0 \\ f=0 \end{cases}$ $L=8$ This now is stable structure too in terms of orbital	→ Energy levels are present. Electrons in different orbital are distinguished in energies depending upon "Quantum Number" → 4 Quantum Numbers 1) Principal (2) Azimuthal (3) Magnetic (3) Spin ⇒ $E = mc^2$ mass → velocity of light → positive electron; positron
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Quantum Numbers :- Energy in terms of quantum numbers

$$E = mc^2$$

i) Principal Quantum Number;

ii) Azimuthal Quantum Number;

iii) Magnetic Quantum Number;

iv) Spin Quantum Number;

Quantum Mechanical Model of the Atom;

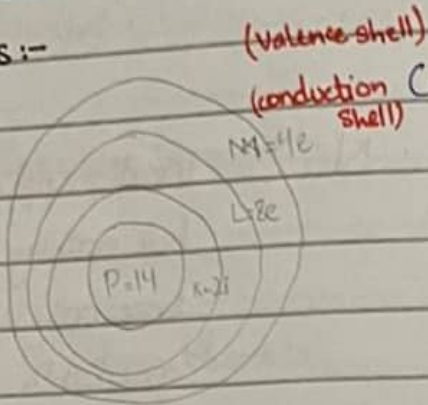
- Heisenberg; uncertainty principle
- Pauli; Exclusion
- Schrodinger; Developed the wave principle
- De Broglie; wave nature of electrons
- Einstein; Relation b/w energy and mass

Copper; ^{29}Cu ; $e = 29$

	K	L	M	N	
Rutherford shell	2	8	18	1	stable structure
Bohr orbital	s=2	s=2, p=6	s=2, p=6, d=10	s=1	→ free electron; reason for conductivity of Cu
	stable	stable	stable		

semiconductor is like insulator in 0°K temp.

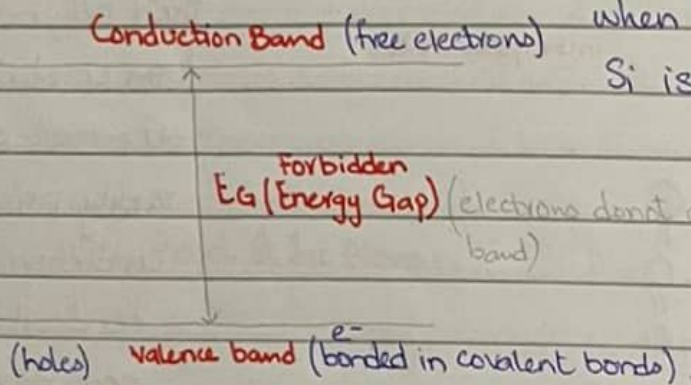
Energy levels :-



Conduction cannot take place due to K and L electrons // can only happen to M shell electrons

Energy Band Diagram :-

Energy (eV)



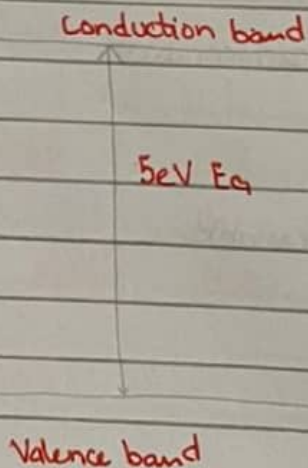
* No energy band in Si until heat is applied and then energy gap happens when e^- jumps. Energy gap in Si is 1.12eV

* E_G of germanium = 0.75eV

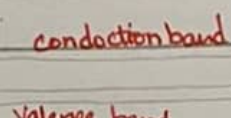
* Germanium have higher conductivity than Si cause lesser E_G

Ionization :- Breaking of ^{covalent} bonds due to temperature and shift from valence to conduction band and results in electrons and holes

Insulator :-



Conductor :-



→ Semiconductor in pure form is called Intrinsic ^{semi} Conductor.

Resistivity:- It is the ability to resist the flow of current

$$\rho = \frac{A \cdot R}{L}$$

A = area of section (cm^2)

L = length of section (cm)

R = Resistance (Ohms)

$$= \frac{\text{cm}^2 \times \text{ohms}}{\text{cm}}$$

$$= \text{cm} \times \text{ohms} / \text{ohms} \times \text{meters}$$

Resistivity value ;

Insulator = $9 \times 10^{11} \text{ ohm} \cdot \text{m}$

Conductor = $1.72 \times 10^{-8} \text{ ohm} \cdot \text{m}$

Semi-conductor = $2 \times 10^3 \text{ ohm} \cdot \text{m}$

754

macro ~~late~~ level (in the lab)

$$\frac{1 \text{ ohm} \times \text{cm}}{\text{cm}^2}$$

$$\frac{1}{\text{ohm}} / \text{mho}$$

— limited to limited

$$\Rightarrow \left(\frac{1}{11} \right)^2 = \left(\frac{1}{11} \right) - \left(\frac{1}{11} \right)$$

$$\frac{1}{\Gamma(BK)} \frac{1}{\Gamma(BK)} = \frac{1}{\Gamma(BK)}$$

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$C = \{L\} \leftarrow \{a\} \leftarrow \{10\} \leftarrow \{7\}$$

2000

1. The first part of the document is a list of names and their corresponding dates. The names are listed in the left column, and the dates are listed in the right column. The names are: John Doe, Jane Smith, and Bob Johnson. The dates are: 1/1/2020, 2/1/2020, and 3/1/2020.

$$\frac{1}{\rho} \frac{d\rho}{dt} = \frac{1}{P} \frac{dP}{dt}$$

(1732)

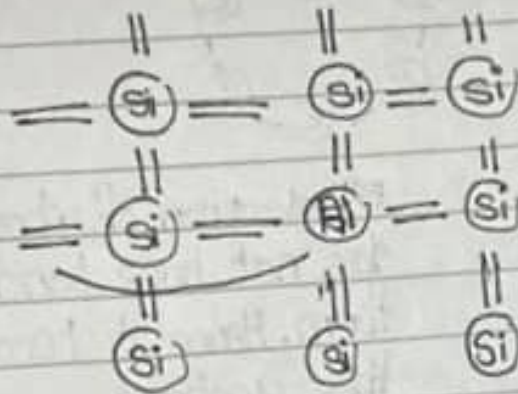
Date: _____

Mon Tue Wed Thu Fri Sat Sun
 ○ ○ ○ ○ ○ ○ ○

• Conduction Band:- e due to ionization + e due to Donor impurity

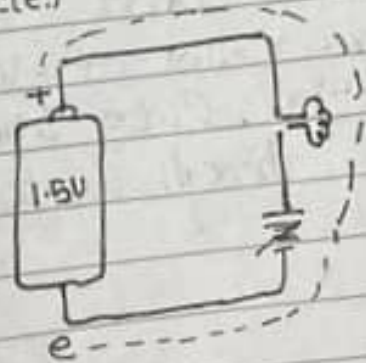
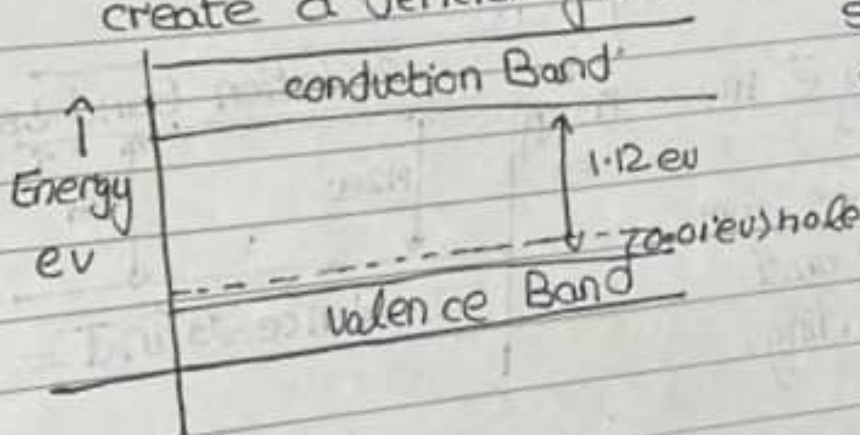
• Valence Band:- holes to ionization
 Extrinsic semiconductor is electrically NEUTRAL.

P-type Extrinsic semi-conductor



¹³ Al	Valence shell		
K	L	M	
2	8	3	
s=2	s=2 p=6	s=3, p=1	

Al (aluminium) is an acceptor type Impurity.
 The extra electron breaks another covalent bond to create a deficiency (ie hole). This is called p-type semiconductor. (The hole is considered as a positive charge particle.)



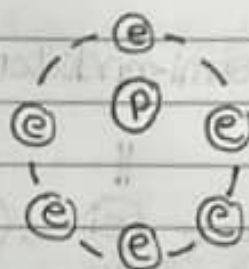
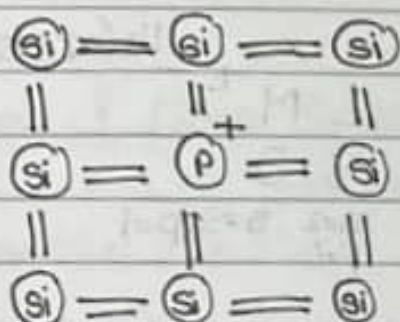
Applied Physics

• Extrinsic semiconductors:-

In this semi-conductor an impurity is added to change its electrical properties. The impurity is called Dopant and the process of adding impurities is called Doping.

$$^{14}\text{Si} \quad K=2 \quad L=8 \quad M=4$$

$$^{15}\text{P} \quad K=2 \quad L=8 \quad M=5$$

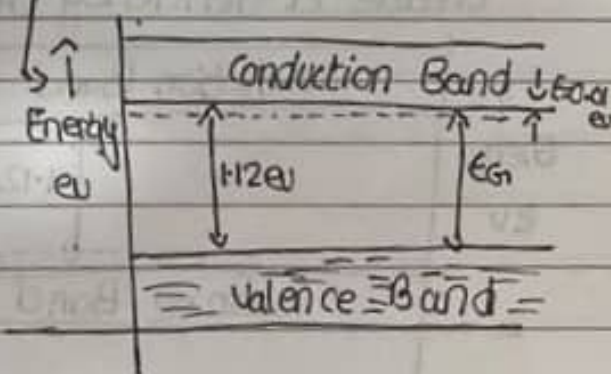


5th electron of phosphorus does not have bond to form. Hence it becomes a free electron.

At room temperature all of the phosphorus 5th electron are in the conduction Band. Phosphorus is a Donor Dopant. This increases the concentration of electrons.

Q:- why do we need so many e^- in conduction Band?

⇒ we need alot of electrons for conduction. later transfer and shifting through ~~conducting~~ bonding.



electron-hole pair

Date: _____

P-type Semiconductor:-

Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun ☐

conduction Band :- e^- due to electron-hole pair.

Valence Band :- holes due to electron-hole + Holes due to acceptor Type Impurity.

P-type material electrically neutral.

Only a redistribution of charge takes place.

Holes the majority carriers.

Electron is the minority carriers } P-type semiconductor.

• Law of Mass Action :-

- n_i concentration of holes/electron in an intrinsic semiconductor.

$$-n_i^2$$

in intrinsic or extrinsic semiconductor.

$$n \times p = n_i^2$$

electron
characteristic
concentration

hole
concentration.

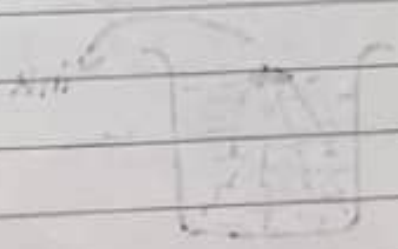
concentration of impurity atoms (phosphorus impurity) $\rightarrow$

$$n_N \cdot p_N = n_i^2 \Rightarrow p_N = \frac{n_i^2}{n_N} = \frac{n_i^2}{N_o}$$

N-type semiconductor.

$$n_p \cdot p_p = n_i^2$$

P-type semiconductor.

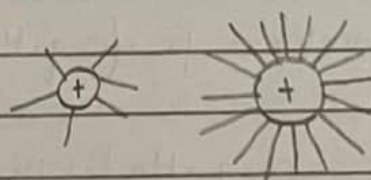


$$\frac{I}{A} = J$$

* cause n (electrons) are in conduction band while p (holes) are in valence band

Drift:-

- Electric field (E) = volts/cm
- Current Density (J) = Amps/cm²
- concentration of (n, p) = 1/cm³
charged particle electron holes
- charge of an e⁻/hole (q) = 1.6 × 10⁻¹⁹ Coulomb



Larger charge more electric field hence

$$I = 5 \text{ mA} \quad \square = 1 \text{ cm}^2 \quad J = \frac{5 \text{ mA}}{1 \text{ cm}^2}$$

$$I = 5 \text{ mA} \quad \square = 5 \text{ cm}^2 \quad J = \frac{5 \text{ mA}}{5 \text{ cm}^2}$$

$$\text{mobility of electrons} = \frac{\text{cm}^2}{\text{volt} \cdot \text{sec}} \quad \text{and holes} \left(\begin{matrix} \mu_n \\ \mu_p \end{matrix} \right)$$

$$\text{conductivity} (\sigma_n, \sigma_p) = \frac{1}{\text{ohm} \cdot \text{cm}}$$

$$\text{Drift velocity} (V_d) = \text{cm/sec}$$

$$\Rightarrow V_d = \mu E$$

$$= \frac{\text{cm}^2}{\text{volt} \cdot \text{sec}} \times \frac{\text{volts}}{\text{cm}}$$

$$= \text{cm/sec}$$

$$\Rightarrow J = n \cdot q \cdot V_d$$

more no. of electrons or charge hence more V_d

$$= \frac{\text{cm}^3}{\text{cm}^3} \cdot \frac{\text{coulomb}}{\text{cm}^3} \cdot \frac{\text{cm}}{\text{sec}}$$

$$= \frac{\text{coulomb}}{\text{cm}^2 \cdot \text{sec}} = \frac{\text{Amp}}{\text{cm}^2}$$

* $\rightarrow \text{coulomb/sec} = \text{Amp}$

$$\Rightarrow J = n \cdot q \cdot V_d \Rightarrow n \cdot q \cdot \mu E \Rightarrow \text{conductivity} (\sigma)$$

$$\Rightarrow J = \sigma \cdot E$$

$\Rightarrow$ N-type material :- (electrons + holes)

$$J_n = q \cdot n \cdot \mu_n \cdot E$$

(current density of electrons)

$$J_{\text{Total}} = (n \cdot q \cdot \mu_n \cdot E) + (p \cdot q \cdot \mu_p \cdot E)$$

$$J_T \Rightarrow \sigma_n \cdot E + \sigma_p \cdot E$$

$$J_p = p \cdot q \cdot \mu_p \cdot E$$

(current density of holes)

$$n \gg p$$

conc. of e⁻ in N type is much greater than conc. of holes
hence $J_T \approx n \cdot q \cdot \mu_n \cdot E$

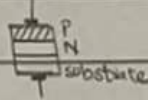
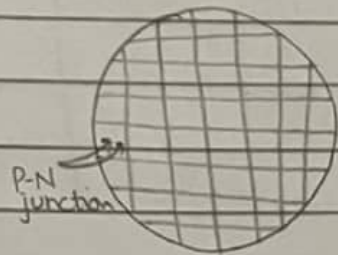
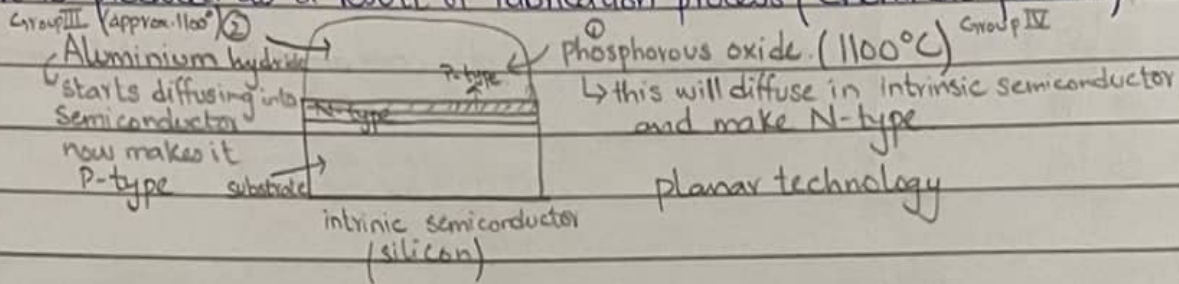
* Movement of electron from N-type to P-type due to diffusion.

* fabrication of P-N junction $\rightarrow$ youtube

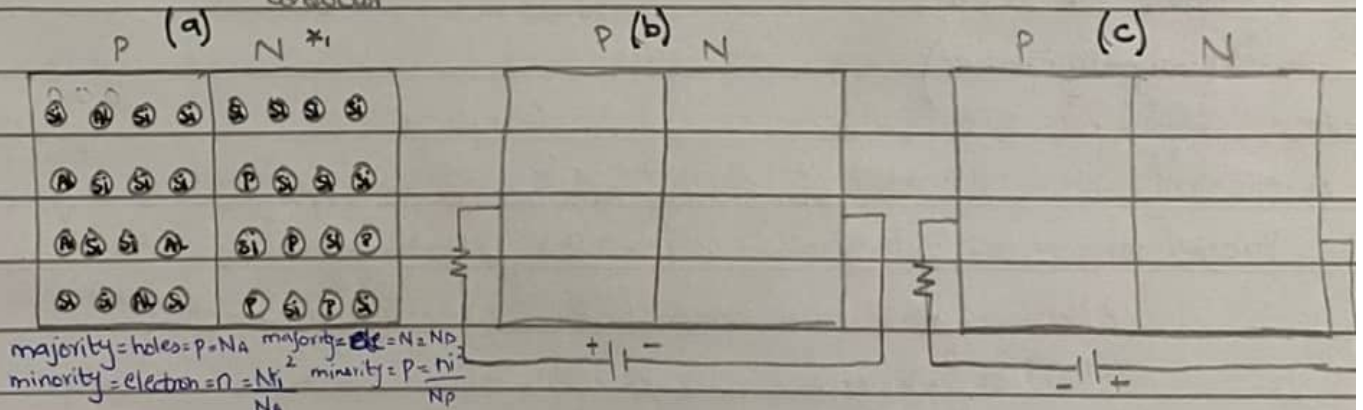
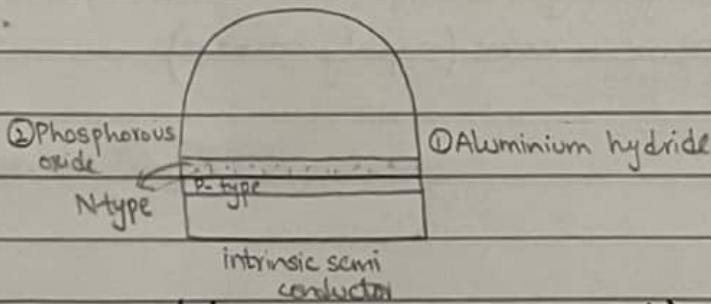
P-N junction :-

Comprises P-type semiconductor joined to N-type semiconductor

It is produced as a result of fabrication process (chemical process)



attach the terminal encapsulate it to give strength



a) when no external is applied, what is the state of subatomic particle.

UNI-Biased P-N junction - Movement of holes from P to N-type due to diffusion. Vice versa for

b) when p type is connected to +ve and N-type to -ve it is called forward bias

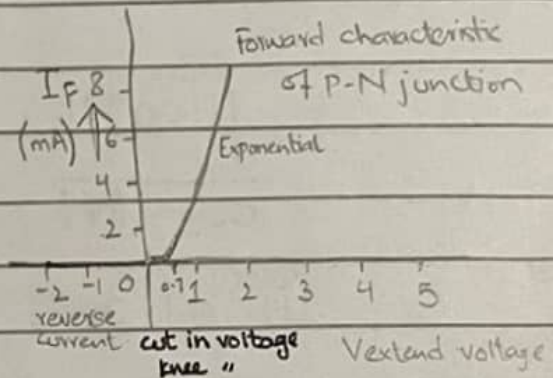
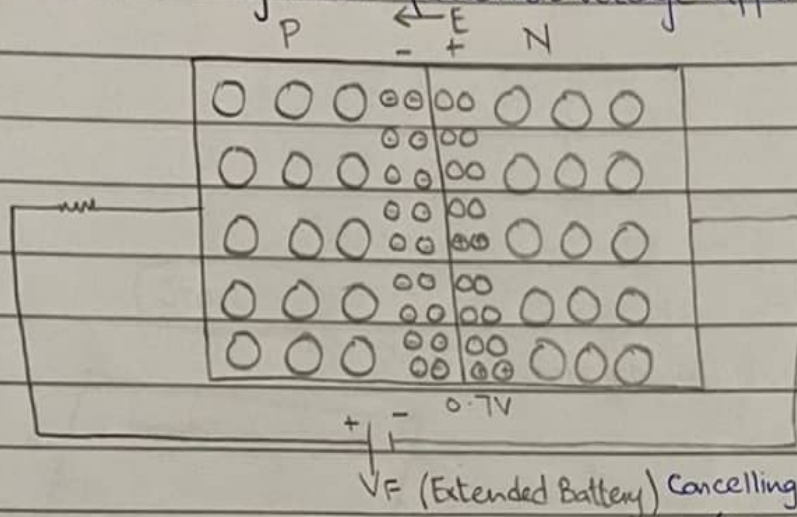
c) " N type " " " " " P-type " " " " reverse biased

P-N junction

Potential ~~difference~~ ^{developed} at junction is called Barrier Potential or Junction Potential
 Equilibrium situation (no net current)

V_{BP}
 (P-N)_i Si (0.7V)
 G (P-N)_{junction} (1.2V-2V)

Forward biased P-N junction :- (external voltage applied; +ve to P and -ve to N-type)

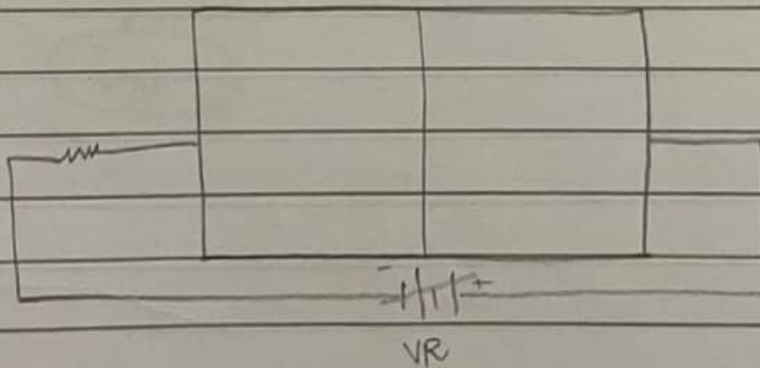


V_F (Extended Battery) Cancelling the effect of Barrier Potential (V_{BP})
 ($V_F - V_{BP}$) current will flow

Majority carrier holes $\rightarrow$ P to N
 " " electrons $\rightarrow$ N to P

This current is forward current I_F

Reversed Biased P-N junction :-



V_R increases the Barrier potential so no current flows across the junction. Junction is said to be reverse biased

$$V_{BP} = 0.7 \text{ volts}$$

$$V_R = 0.1 \text{ volts}$$

$$\text{Total Barrier potential} = 1.7 \text{ V}$$

However minority carriers will be able to cross the junction.

$$298^{\circ}\text{K} \quad 25^{\circ}\text{C}$$

$$273^{\circ}\text{K} \quad 0^{\circ}\text{C}$$

$$0^{\circ}\text{K}$$

n-type material

$$n \approx N_D$$

$$p = \frac{n_i^2}{N_D}$$

p-type semiconductor

$$p \approx N_A$$

$$n = \frac{n_i^2}{N_A}$$

$$* \text{ Volt} = \frac{\text{Joules}}{\text{C}}$$

Q4. $n_i = 1.5 \times 10^{10} / \text{cm}^3$

$n = 5 \times 10^{14} / \text{cm}^3$ (suppose it is n-type)

$p = ?$

$$* p = \frac{n_i^2}{N_D} = \frac{(1.5 \times 10^{10})^2}{(5 \times 10^{14})} \cdot \frac{1}{(\text{cm}^3)^2} \cdot \frac{\text{cm}^3}{1}$$

$$= \frac{2.25 \times 10^{20}}{5 \times 10^{14}} \cdot \frac{1}{\text{cm}^3}$$

$$= 0.462 \times 10^6 \cdot \frac{1}{\text{cm}^3}$$

$$= 0.462 \text{E}6 \cdot \frac{1}{\text{cm}^3}$$

Q5. $n \approx \frac{n_i^2}{N_A}$

Q6. $4 \times 10^{19} / \text{cm}^3$ a) Silicon atom (b) holes (c) dopant atoms (d) valence electron

$$n_i^2 \approx 10^{10} / \text{cm}^3$$

$p = "$ (b not possible)

$n = "$ $n \times p = n_i^2$

^{14}Si (14 electron) (rule out a)

K. L. M^x (rule out d)

2 8 4

Q7. $N_D = 2.25 \times 10^{15} \text{ atoms} / \text{cm}^3$

$n_i = 1.5 \times 10^{10} / \text{cm}^3$

$n = ?$

$p = ?$

$$p = \frac{(1.5 \times 10^{10})^2}{(2.25 \times 10^{15})} \cdot \frac{1}{(\text{cm}^3)^2} \cdot \frac{\text{cm}^3}{1}$$

$$= \frac{2.25 \times 10^{20}}{2.25 \times 10^{15}} \cdot \frac{1}{\text{cm}^3}$$

$$= 1 \times 10^5 \cdot \frac{1}{\text{cm}^3}$$

Q8. Einstein Relationship; $D_p = \mu_p \frac{k_B T}{e}$

$D_p =$ diffusion constant of holes

$$D_p = \frac{500 \times (1.38 \times 10^{-23}) \times 300^{\circ}\text{K}}{1.60 \times 10^{-19} \text{ volt} \cdot \text{sec}} \cdot \frac{\text{Joule}}{\text{C}} \cdot \frac{1}{\text{C}}$$

μ_p k_B T e

$$12.9375 \cdot \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \cdot \frac{\text{V}}{\text{C}} \cdot \frac{1}{\text{C}}$$

$$12.93 \frac{\text{cm}^2}{\text{s}}$$

0.142
0.64

$\frac{\text{Coulomb}}{\text{sec}} = \text{Ampere}$

Q9 = N-type (donor)

$$N_D = 10^{16} / \text{cm}^3$$

$$\mu_n = 1200 \text{ cm}^2 / \text{V} \cdot \text{sec}$$

$$\mu_p = 400 \text{ cm}^2 / \text{V} \cdot \text{sec}$$

$$C = 1.6 \times 10^{-19} \text{ C}$$

$$\sigma = ?$$

$$\sigma = n \times q \times \mu_n$$

$$= 10^{16} \times (1.6 \times 10^{-19}) \times (1200) \quad \frac{1}{\text{cm}^3} \cdot \frac{\text{C}}{\text{V} \cdot \text{sec}} \cdot \frac{\text{cm}^2}{\text{V} \cdot \text{sec}}$$

$$= 1.92 \quad \frac{\text{Ampere}}{\text{Volts} \cdot \text{cm}}$$

$$= 1.92 \quad \frac{1}{\text{ohm} \cdot \text{cm}}$$

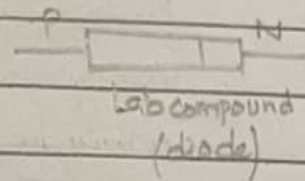
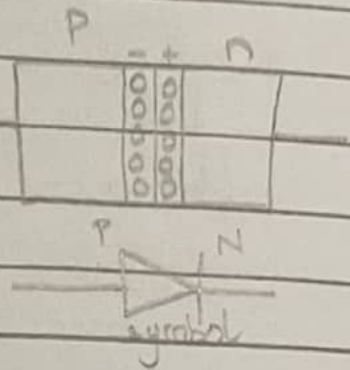
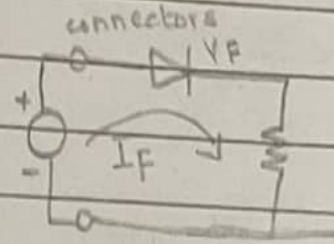
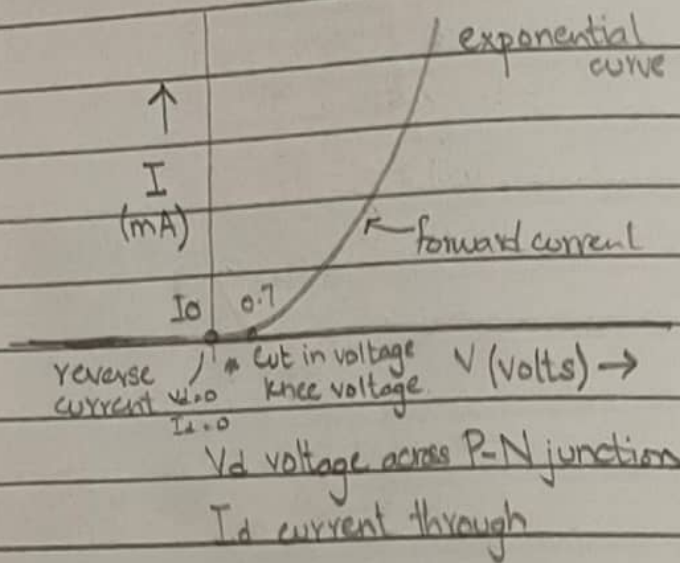
$$p = \frac{n_i^2}{n_D} = \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^4$$

$$\sigma_p = 2.25 \times 10^4 \times 1.6 \times 10^{-19} \times 400$$

$$= 1.44 \times 10^{-12} \quad (\text{can be ignored})$$

$$\begin{aligned} n + p &= n_i \\ n_D + p &= n_i \\ p &= \frac{n_i^2}{n_D} \\ n + p &= n_i \end{aligned}$$

Characteristics of P-N junction:-



$$i_d = I_0 e^{\frac{V_d}{nV_T}}$$

$$i_d = I_0 [e^{\frac{V_d}{nV_T}} - 1] \quad (\text{General equation})$$

i_d = diode current (through the diode) - variable

V_d = " voltage (across " ") - "

I_0 = reverse saturation current - constant

n = constant [$n=1$ for Ge and $n=2$ for Si diode]

$V_T = \frac{kT}{q}$ = Thermal voltage

To apply general equation

$$V_d = 0$$

$$i_d = I_0 [e^0 - 1] \quad V_d = 0, i_d = 0$$

$$i_d = I_0 [1 - 1] = 0$$

$$V_d = 0.3 \text{ volts}$$

$$i_d = I_0 [e^{\frac{0.3}{2(0.026)}} - 1]$$

$$= I_0 [e^{\frac{0.3}{0.052}} - 1]$$

$$= I_0 [e^{5.769} - 1]$$

$$= I_0 [320 - 1]$$

$$= I_0 [319]$$

$$= \mu\text{Amp} [319] \approx \mu\text{Amps} \approx 0$$

$$V_d = 1 \text{ volt}$$

$$i_d = I_0 [e^{\frac{1}{2 \times 0.026}} - 1]$$

$$= I_0 [e^{19.2} - 1]$$

$$= I_0 [2.2 \times 10^8 - 1]$$

$$= (1 \times 10^{-6}) (2.2 \times 10^8)$$

$$= 2.2 \times 10^2$$

$$= 220 \text{ Amps.}$$

$$\text{Volts} = \frac{\text{Joules}}{\text{coulomb}}$$

$$\text{coulomb} = \frac{\text{Joules}}{\text{volts}}$$

Numericals :-

Q2 = $E = 10 \frac{\text{V}}{\text{cm}}$

$d = 1 \text{ cm}$

$t = 20 \text{ Hsec}$

velocity = $\frac{\text{distance}}{\text{time}} = \frac{100 \text{ cm}}{20 \times 10^{-6} \text{ sec}}$

$V = 0.05 \times 10^6 \frac{\text{cm}}{\text{sec}}$

$V = \mu E$

$\mu = \frac{V}{E} = \frac{0.05 \times 10^6}{10} \frac{\text{cm} \cdot \text{cm}}{\text{sec} \cdot \text{V}}$

$= 5 \times 10^3 \frac{\text{cm}^2}{\text{volt} \cdot \text{sec}}$

Q3 = Find the value of q/KT of room temp. (300°K)

$q = 1.6 \times 10^{-19} \text{ C}$

$K = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$

$\frac{1.6 \times 10^{-19}}{1.38 \times 10^{-23} \cdot 300} \frac{\text{C} \cdot \text{K}}{\text{J} \cdot \text{K}}$

$0.003864 \times 10^4 \frac{\text{C}}{\text{J}} \rightarrow V = \frac{\text{J} \cdot \text{C}}{\text{J}}$

$38.64 \times \frac{\text{J} \cdot \text{C}}{\text{V} \cdot \text{J}}$

$38.64 / \text{volts} \Rightarrow 38.64 \text{ volt}^{-1}$

$$V_d = 0.75$$

$$I_d = I_0 \left[e^{\frac{0.75}{2 \times 0.026}} - 1 \right]$$

$$= I_0 \left[e^{\frac{0.75}{0.052}} - 1 \right]$$

$$= I_0 \left[e^{14.4} - 1 \right]$$

$$= 1 \times 10^{-6} [1.83 \times 10^6 - 1]$$

$$= 1.83 - 1$$

$$V_d = -2 \text{ volts}$$

$$I_d = I_0 \left[e^{\frac{-2}{2 \times 0.026}} - 1 \right]$$

$$= I_0 \left[e^{-\frac{38.4}{0.052}} - 1 \right]$$

$$= I_0 \left[e^{-\frac{1}{38.4}} - 1 \right]$$

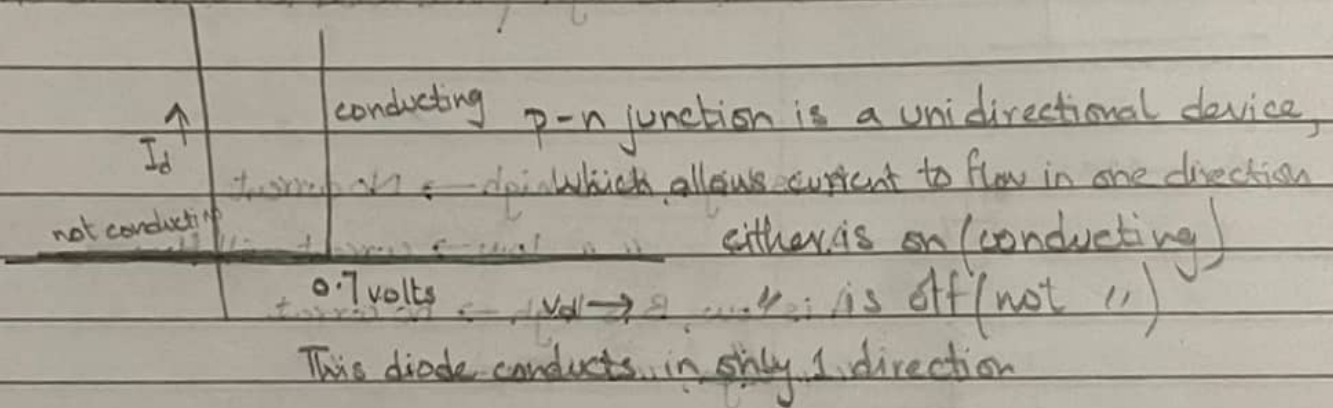
$$= I_0 [0.71 - 1]$$

$$= -0.29 I_0$$

$$= -0.29 \times 10^{-6} \text{ A}$$

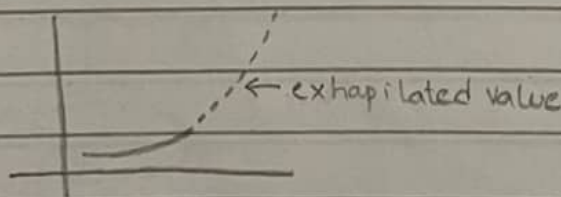
Current is 1.83 Amps

(It is not a diode, it is a resistor)



Application of P-N junction :-

- 1) Implementation of Boolean Arithmetic (Boole 1850)
Base-2 Number system (only 0 and 1)



After invention of P-N junction, the Boolean Arithmetic was put into practice

- 2) Any Complex process can be broken down into binary (Yes/No) decision, and result can be obtained after series of such decisions

P-N junction diode can be used in hardware design of binary decisions.

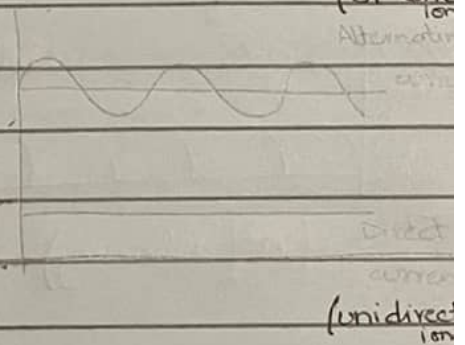
Applications of P-N junction Diode

- 1) Rectification (AC to DC voltage)
- 2) Stabilization (of DC voltage)

Generation of electricity is done by Alternating current

- Economic reason $\rightarrow$ cheaper to transmit electrical energy to AC. Use of transformer in AC (not possible in DC)

$$EMF = N \frac{d\phi}{dt}$$

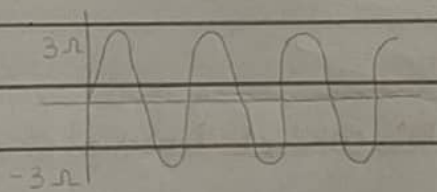
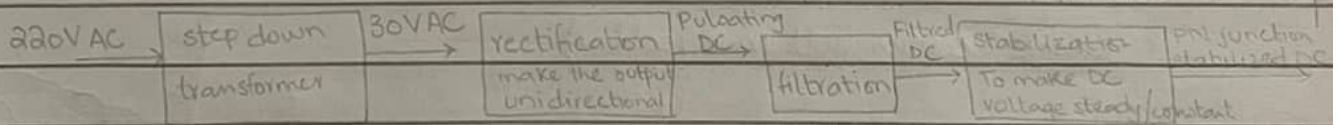


- Electrical devices (based on semi conductors) require DC power - low DC voltage (upto 30V DC)

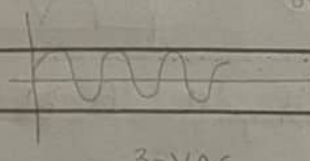
- Hence need to convert from AC to DC

Steps in conversion from AC to DC:-

PN junction diode

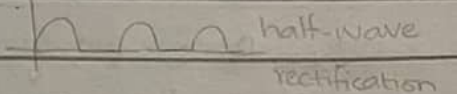


220V AC
50 Hz

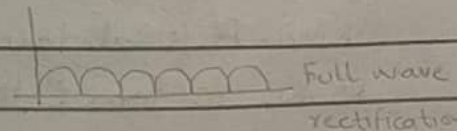


30V AC

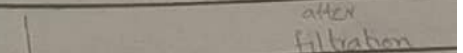
50 Hz



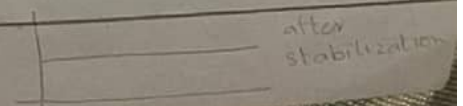
half-wave rectification



Full wave rectification



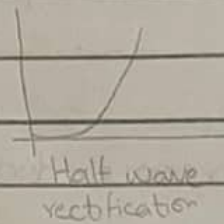
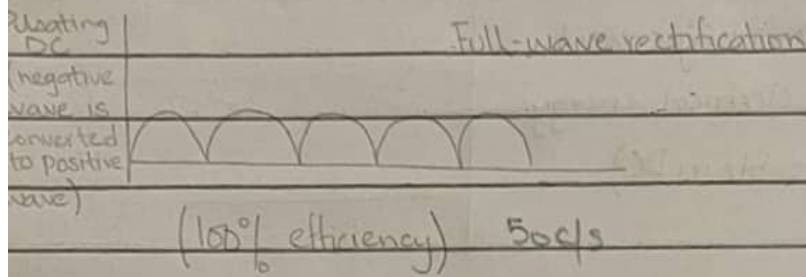
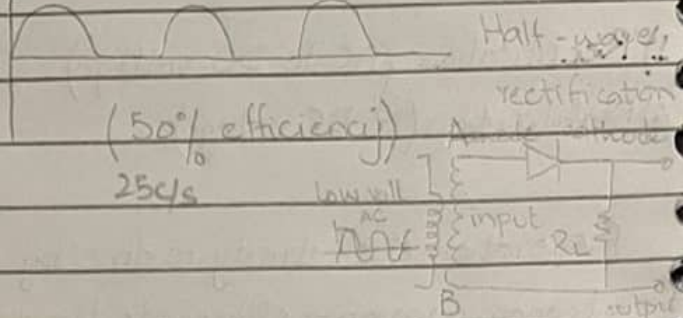
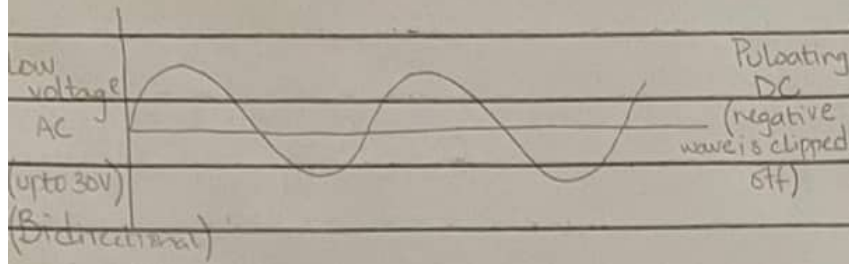
after filtration



after stabilization

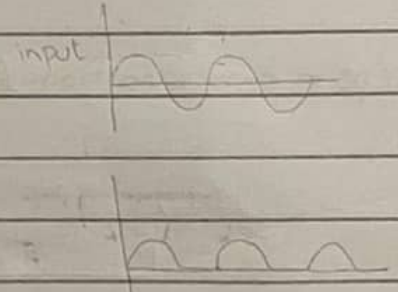
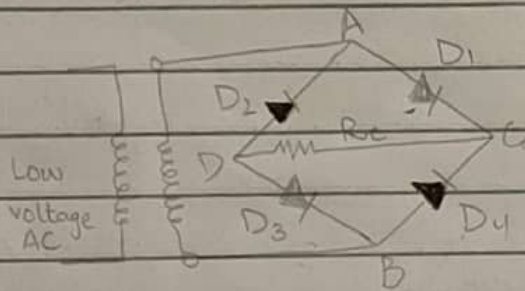
Kirchoff voltage law
 $\sum \text{voltage-rise} = \sum \text{voltage fall}$
 in a closed circuit.

P-N junction is a voltage controlled device.

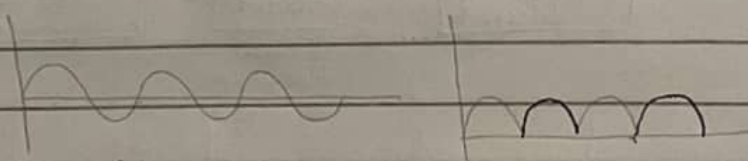


When A is at higher voltage than B (in the +ve wave), diode conductor.

When A is at lower voltage than B (in -ve wave), diode is off.



Bridge Rectifier (full wave rectifier)

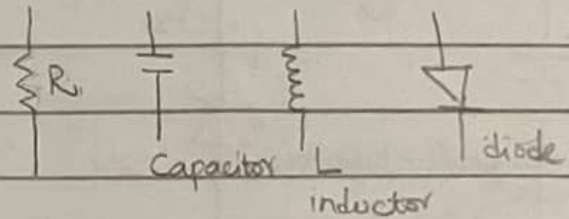


When A is at higher voltage than B, D1 and D2 conduct. Direction of anode in R_L is from C to D

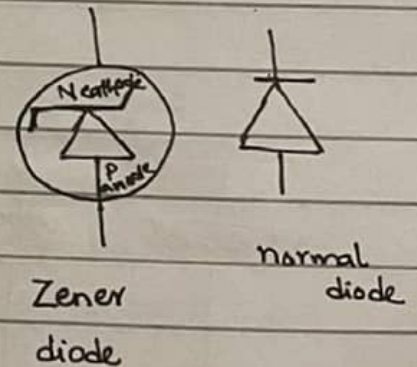
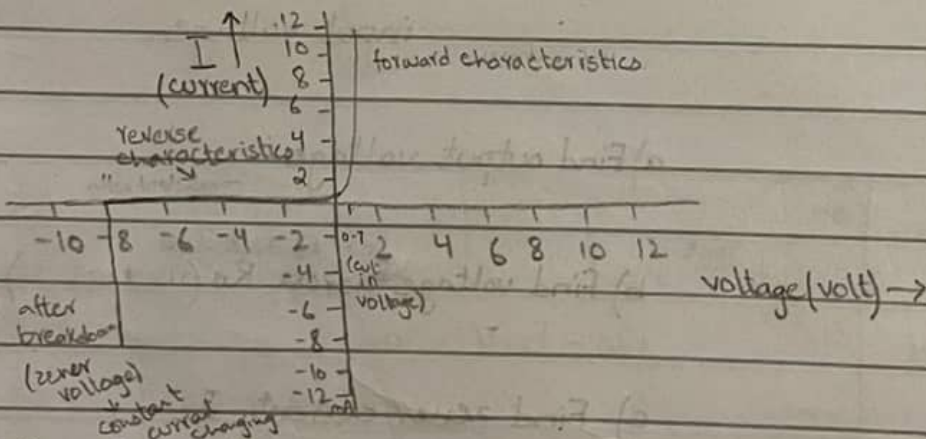
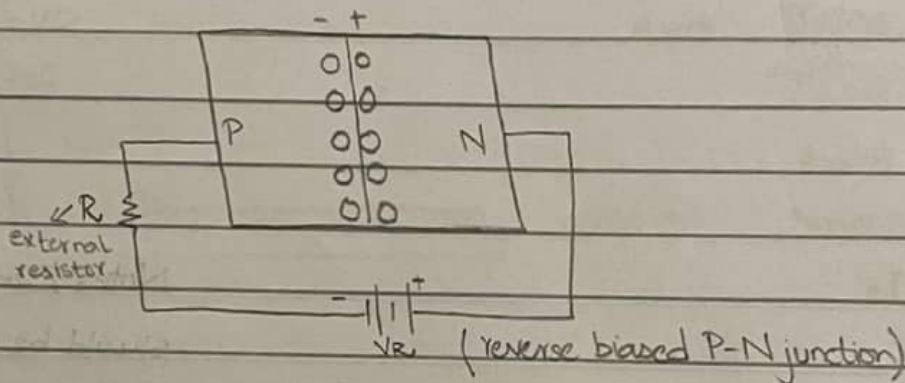
When A is at low voltage than B, D4 and D3 conduct and direction of R_L is from C to D.

Application of P-N junction

- 1) Rectification
- 2) Stabilization (Zener diode)



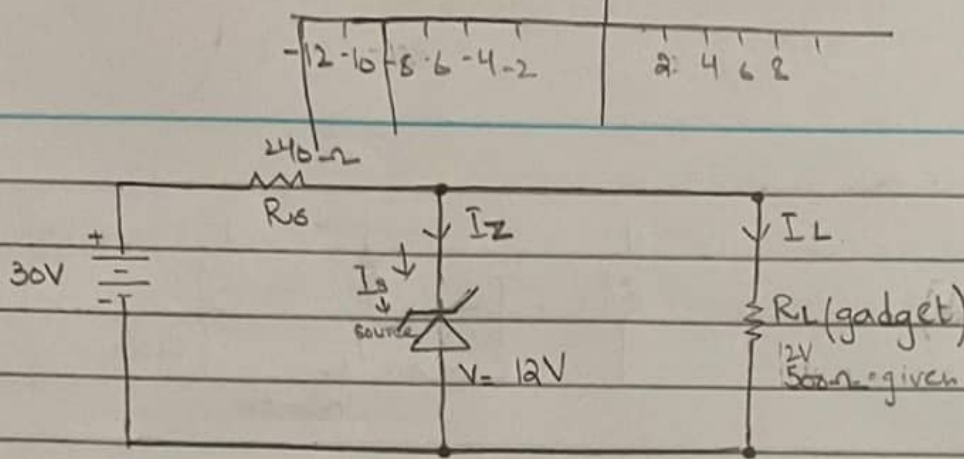
Zener diode is a P-N junction operating in the Reverse Biased Region. In its fabrication it is heavily doped and design to have different breakdown voltages.



When the electrons are pulled out of covalent bonds due to high -ve voltage, Zener breakdown occurs and a large -ve current flows, which is only controlled by the external resistor. (sort of damage the diode). By varying the doping levels ^{more} different breakdown ^{lesser} voltages can be achieved (ie. -8V, -10V, -12V.... -30V).

Ex-28

$$1.35 \text{ mV} = 1.35 \text{ V}$$



a) voltage across load (R_L)
voltage is parallel so
same V in zener
and gadget = 12V.

c) current through zener; I_Z

$$V = IR$$

$$18 = I_s \times 240$$

$$I_s = \frac{18}{240} = \frac{18}{0.24K\Omega} = 75mA$$

$$I_L = \frac{12}{0.5K\Omega} = 24mA$$

$$75 = 24 + I_Z$$

$$75 - 24 = I_Z$$

$$51mA = I_Z$$

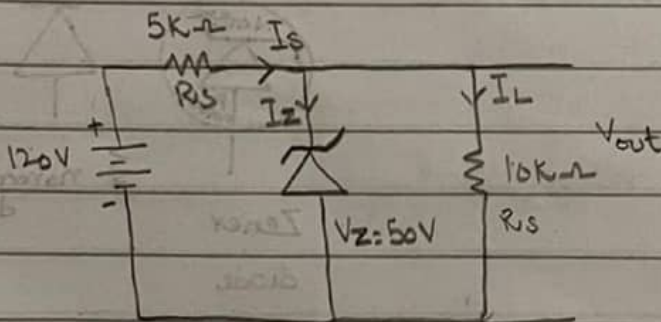
b) voltage drop across R_s (apply Kirchhoff's law)

30V = voltage rise

12V = drop

30 - 12 = 18 voltage

Note: Zener voltage
should be lower than
input voltage.



a) Find output voltage

b) Find voltage across R_s (vs Kirchhoff)

$$120 - 50V = 70V$$

c) Find zener current I_Z

$$V = IR$$

$$70 = I_s \times 5K\Omega$$

$$I_s = 14mA$$

$$I_s = I_Z + I_L$$

$$14 = I_Z + 5$$

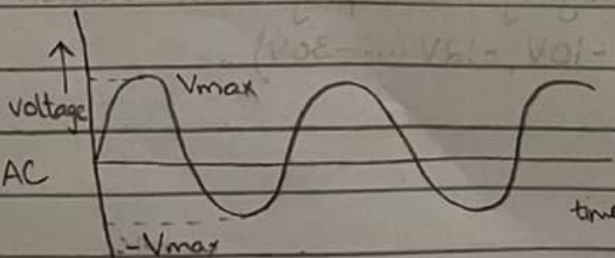
$$14 - 5 = I_Z$$

$$9mA = I_Z$$

Numerical on Rectifier

Due to generator
action, the

rate of change
of magnetic
flux causes
the sin wave



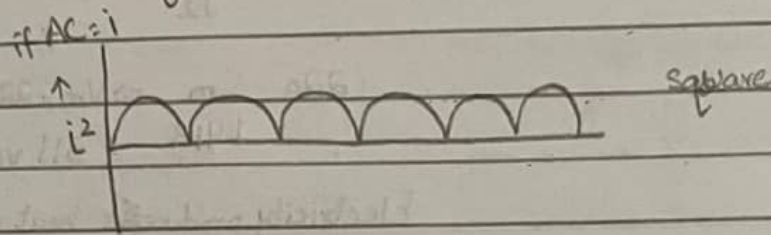
50 cycles/second change

Avg value of AC = 0

$$V(t) = V_m \sin 2\pi ft$$

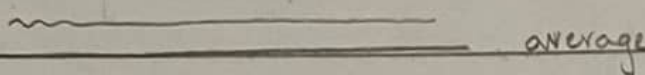
$$\Rightarrow V_m \sin 100\pi t$$

To measure things in AC change to positive only



square root - Average - square
RMS

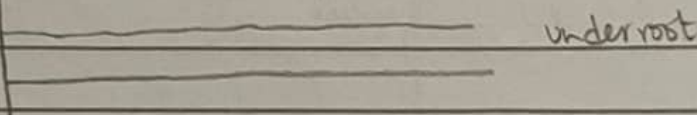
Average of i^2
or mean.



$\Rightarrow$ So for AC we use
root-mean-square value
 V_{RMS} or I_{RMS}

$$I_{RMS} = \frac{I_{M \rightarrow \max}}{\sqrt{2}}$$

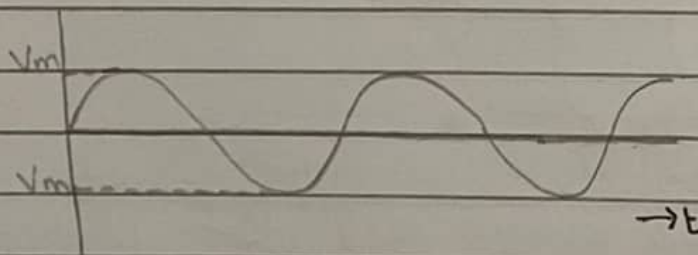
Take $\sqrt{\quad}$ to
get average of
 i



$$V_{RMS} = \frac{V_{M \rightarrow \max}}{\sqrt{2}}$$

Numerical on Rectifiers :-

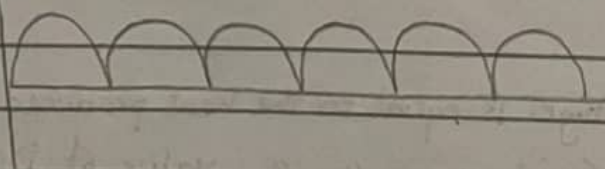
AC
voltage
 $v(t)$



$$\begin{aligned} v(t) &= V_m \sin 2\pi f t \\ &= V_m \sin 2\pi \cdot 50 \cdot t \\ &= V_m \sin 100\pi t \text{ volt} \end{aligned}$$

Avg value = 0

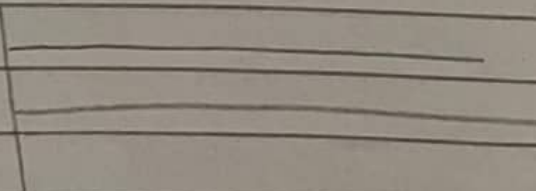
Square
of $v(t)$



Mean(Avg)

of square value

of $v(t)$



square root of mean of the square

RMS value

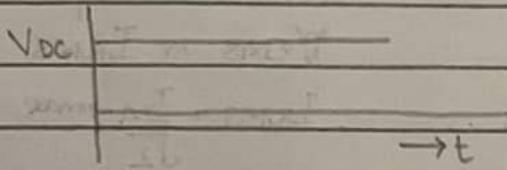
Frequency of electrical supply = $50 \times 12 = 600 \text{ Hz}$

$$V_{\text{RMS}} = \frac{V_m}{\sqrt{2}}$$

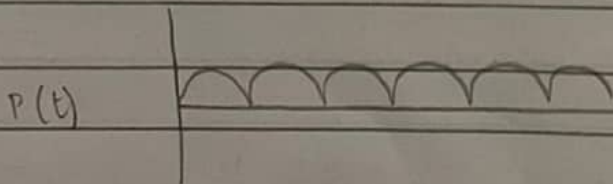
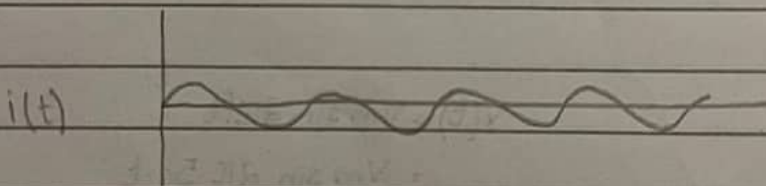
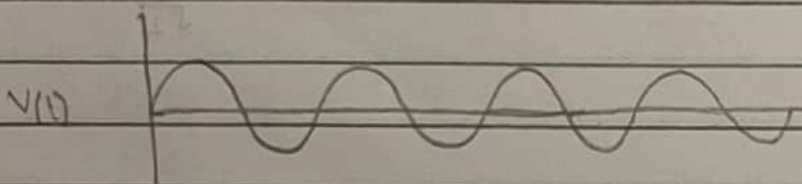
$$220 = \frac{V_m}{1.414} \Rightarrow V_m = 220 \times 1.414 = 311 \text{ volts}$$

Electricity produces heat energy

How to compare heat produced by AC with the heat produced by DC?



$$\text{power} = \frac{(V_{\text{DC}})^2}{R} = (V_{\text{DC}})(I_{\text{DC}}) = (I_{\text{DC}})^2 \times R$$

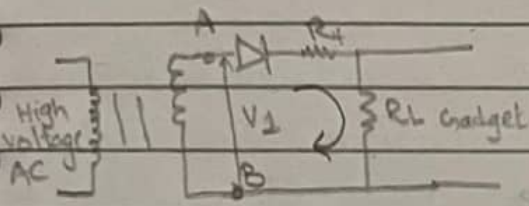


Average power can be determined

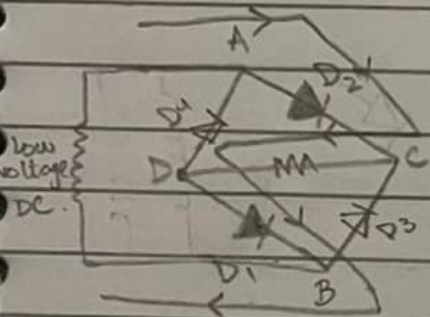
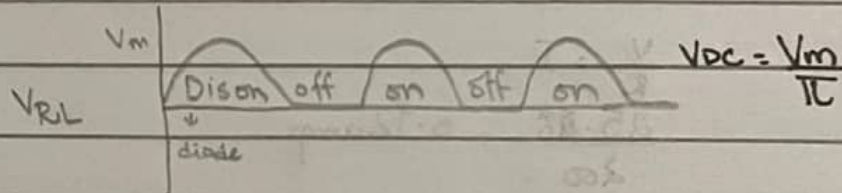
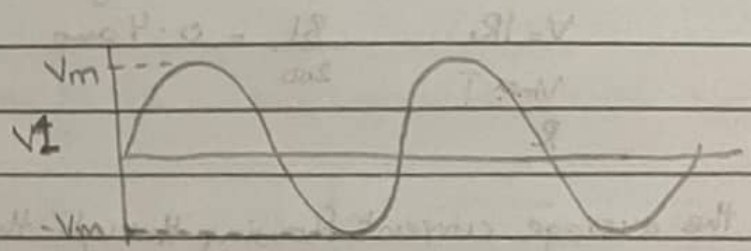
If the heat produced by an AC voltage is equal to the heat produced by a DC voltage, then the RMS value of AC is " " " value of DC.

Average value = DC value = mean value

Transformer:-

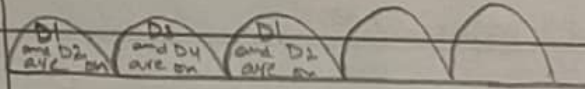
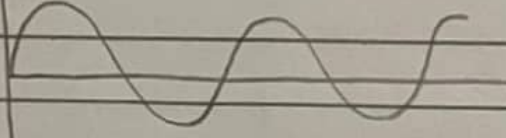


Half wave Rectifier



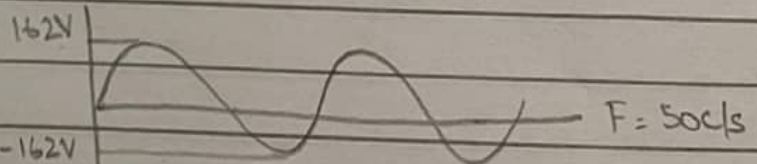
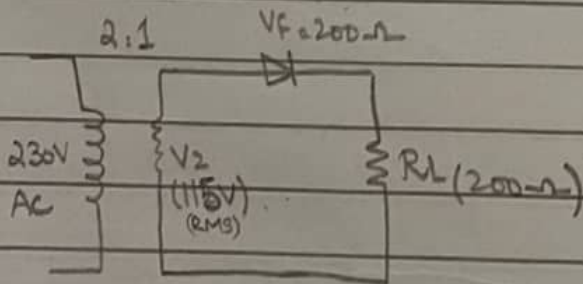
Voltage across the resistor

Input voltage



$$V_{DC} = \frac{2V_m}{\pi}$$

Q1.



$$V_m = \sqrt{2} \times 115V$$

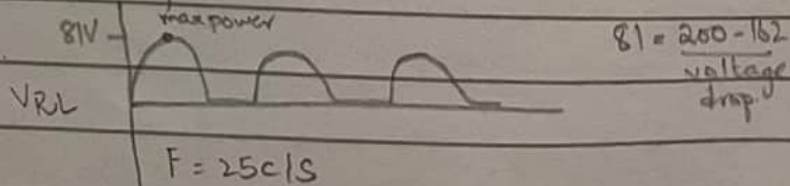
$$= 1.414 \times 115 = 162V \text{ and } -162V$$

Find max power and avg power across load ($R_L = 200\Omega$)

$$P_{\max} = \frac{(V_{\max})^2}{R_L} = \frac{(81)^2}{200} = 32.8 \text{ Watts}$$

$$Avg = V_{DC} = \frac{V_m}{\pi} = \frac{81}{\pi} = 25.8 \text{ volts}$$

$$Avg \text{ Power} = \frac{(25.8)^2}{200} = 3.32 \text{ Watts}$$



Find the max amount amount flowing through the resistor?

$$V = IR \quad \frac{81}{200} = 0.4 \text{ amp.}$$
$$\frac{V_{\max}}{R} = I$$

Find the average current flowing through the resistor?

$$V = IR$$

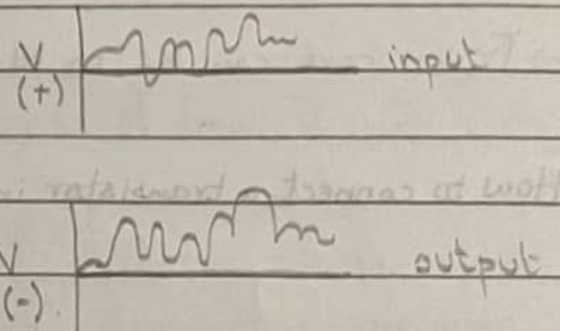
$$\frac{V}{R} = I$$

$$\frac{25.38}{200} = 0.12 \text{ amp}$$

Transistor:-

Important function of transistors are:

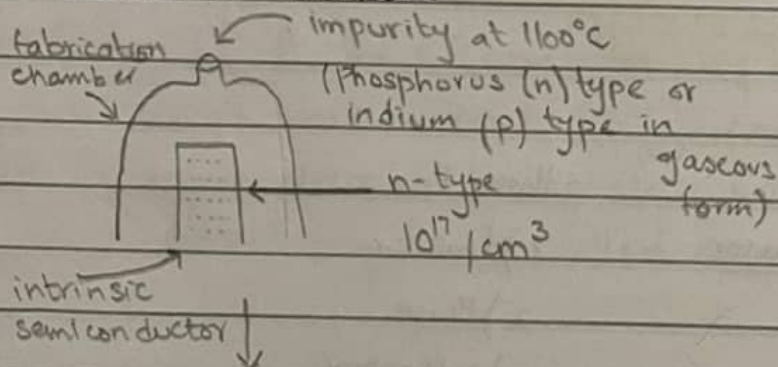
- Amplification:** Increases the magnitude of the output signal as exact replica of the input signal.
- Switching:** To connect and disconnect part of an electric circuit from another part at a high speed.



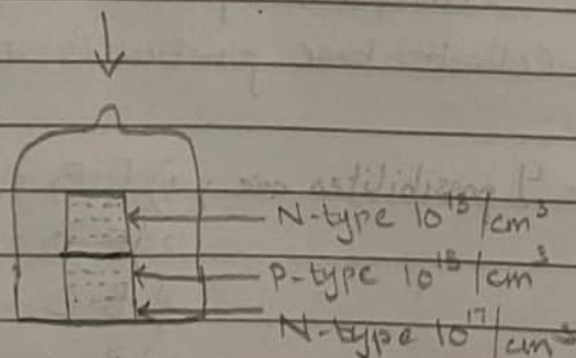
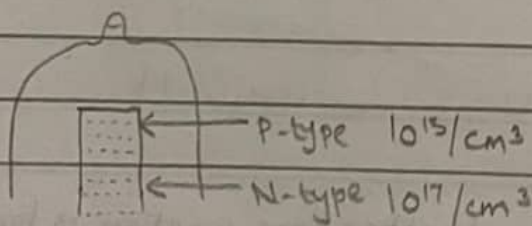
In (a) the transistor operates in active region and in (b) it is in cut off region.

↑ amp or magnitude ↑
waveform is the same.

Fabrication of transistors

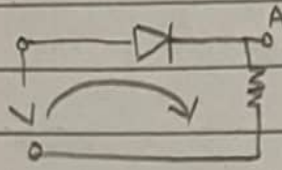


- conc. of impurity
- time duration while exposed to impurity
- determine the type of semiconductor (n or p type)



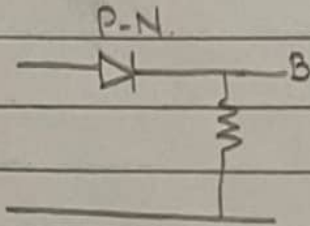
50 μm		← n-type
1 μm		← p-type
5 μm		← n-type

Characteristics of P-N junction :-

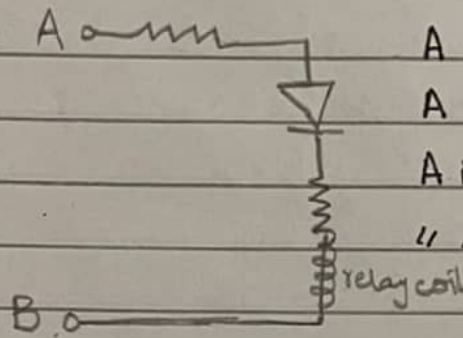


If coin is present, V is applied, current will flow A voltage will become high.

If coin not present, V is not applied, current will not flow $\therefore$ Voltage A becomes low.



If trolley switch is ON, Voltage is applied current will flow and B will be high (vice versa for switch is off).



A is high, B is high $\rightarrow$ No current

A " " " " low $\rightarrow$ current will flow coin there

A is low, B is high $\rightarrow$ no current

" " " " low $\rightarrow$ " "

Base - 2 Number system (only 0 and 1)
Implementation of Boolean Arithmetic (Chapter 1.2.2)

After invention of P-N junction, the Boolean Arithmetic was put into practice

Any complex process can be broken down into binary (yes/no) decision and

result can be obtained after series of such decisions

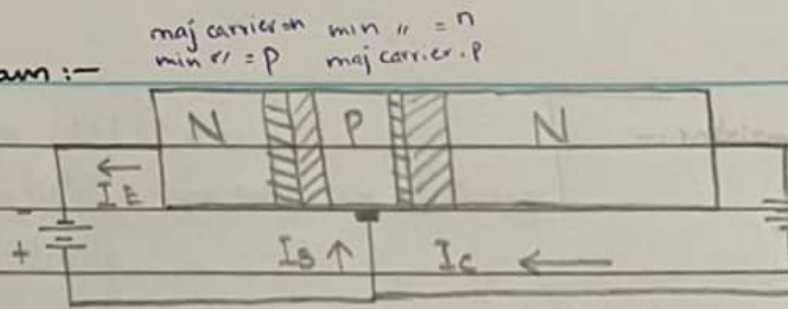
P-N junction diode can be used in logic

design of binary decisions

Fabrication Diagram :-

$$I_E = I_C + I_B$$

$$100 = 95 + 5$$



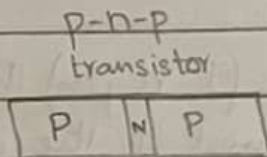
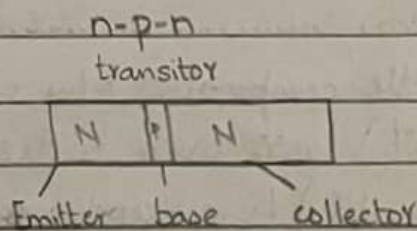
- Electrons are driven from emitters to base because of forward bias E-B junction
- In the base (P-region) there is very little combination b/w electrons and holes and most of the electrons (95%) cross over to the collector side (majority because of the small width) (we keep base's width small to avoid recombination of holes of e^- hence they are attracted across as soon as they reach base)
- Only 5% of the e^- combine w/ holes region
- A small base current (5%) controls a much larger collector
- Hence, transistor is a current controlled device, and the process of I_B controlling I_C is called amplification

*current only flows in active and saturation mode

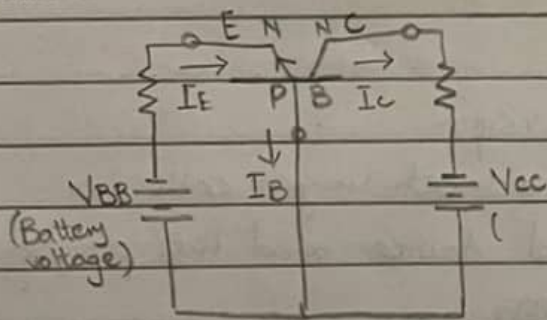
light charge in fabrication
For different purposes there are different transistors

Bipolar Junction Transistor:-

- ~~electrons~~ electrons
- holes
- current controlled device



Fabrication circuit:-



$$I_E = I_B + I_C$$

very small

$$\beta(\text{gain}) = \frac{I_C}{I_B} \quad \alpha(\text{gain}) = \frac{I_C}{I_E}$$

Common base transistor circuit

- β in terms of α

$$I_E = I_B + I_C \Rightarrow \frac{I_E}{I_B} = 1 + \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E} \Rightarrow \frac{I_C}{I_B} = \frac{\alpha I_E}{I_B} = \alpha \frac{I_B + I_C}{I_B} = \alpha (1 + \frac{I_C}{I_B})$$

- α in terms of β

$$\beta - \alpha \beta = \alpha$$

$$\beta(1 - \alpha) = \alpha$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \alpha + \alpha \beta$$

$$\beta = \alpha(1 + \beta)$$

$$\alpha = \frac{\beta}{1 + \beta}$$

Emitter Base Junction

- Forward Bias

- " "

- Reverse "

- Reverse "

circuit does not work in this situation

Collector Base Junction

- Reverse Bias

- Forward "

- Reverse Bias

- Forward "

Mode

* works as amplifier
- Active

- Saturation voltage most

- Cut off

- inverted not used

acts as switch

Note the doping, less is width of depletion region.

drain current (I_D)

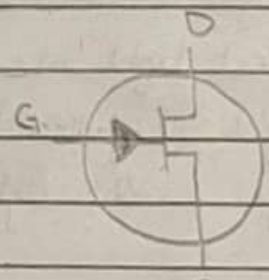
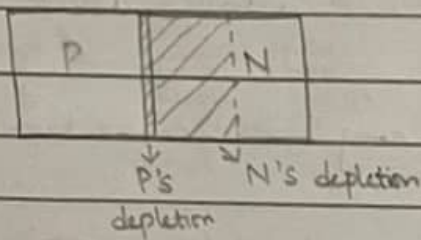
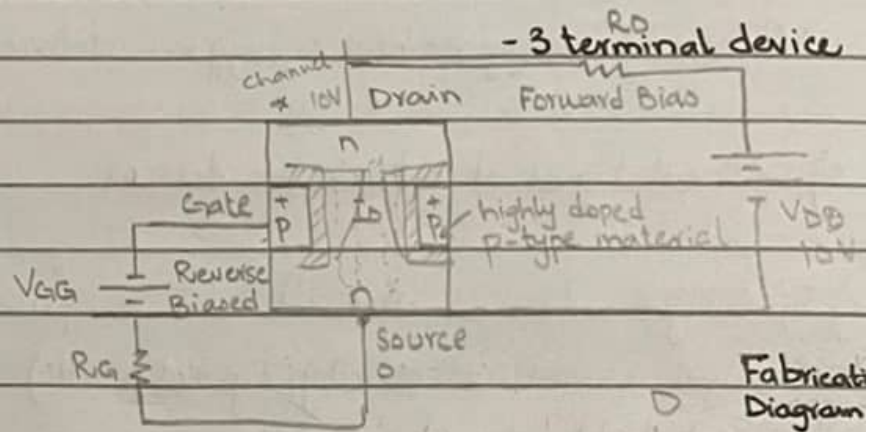
Uni Junction Transistor (UJT) :-

Only one type of particle either holes or electrons

- Voltage controlled device

- N channel UJT

- P channel UJT

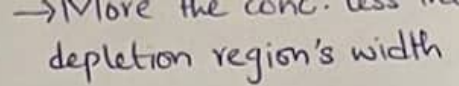


circuit symbol

* voltage drop in channel

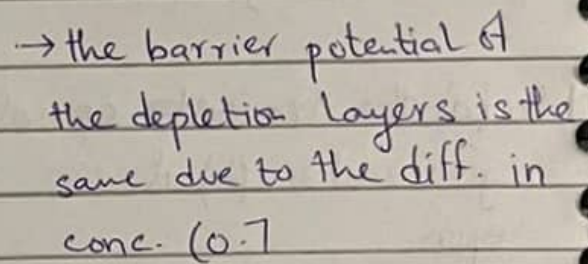
- The higher the voltage, the greater is the depletion region, invade to produce the barrier potential.

Increasing the drain voltage will increase the current, and also make the channel narrow. At a certain point current becomes constant. It is called Pinch off voltage.

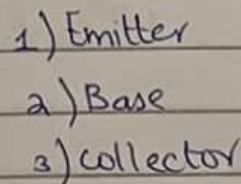


→ Transistors can be in N-P-N or PNP forms

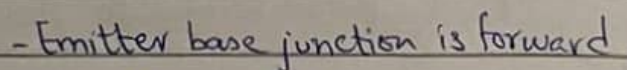
metal contact



- three terminal device



Peconnected to me (forward)



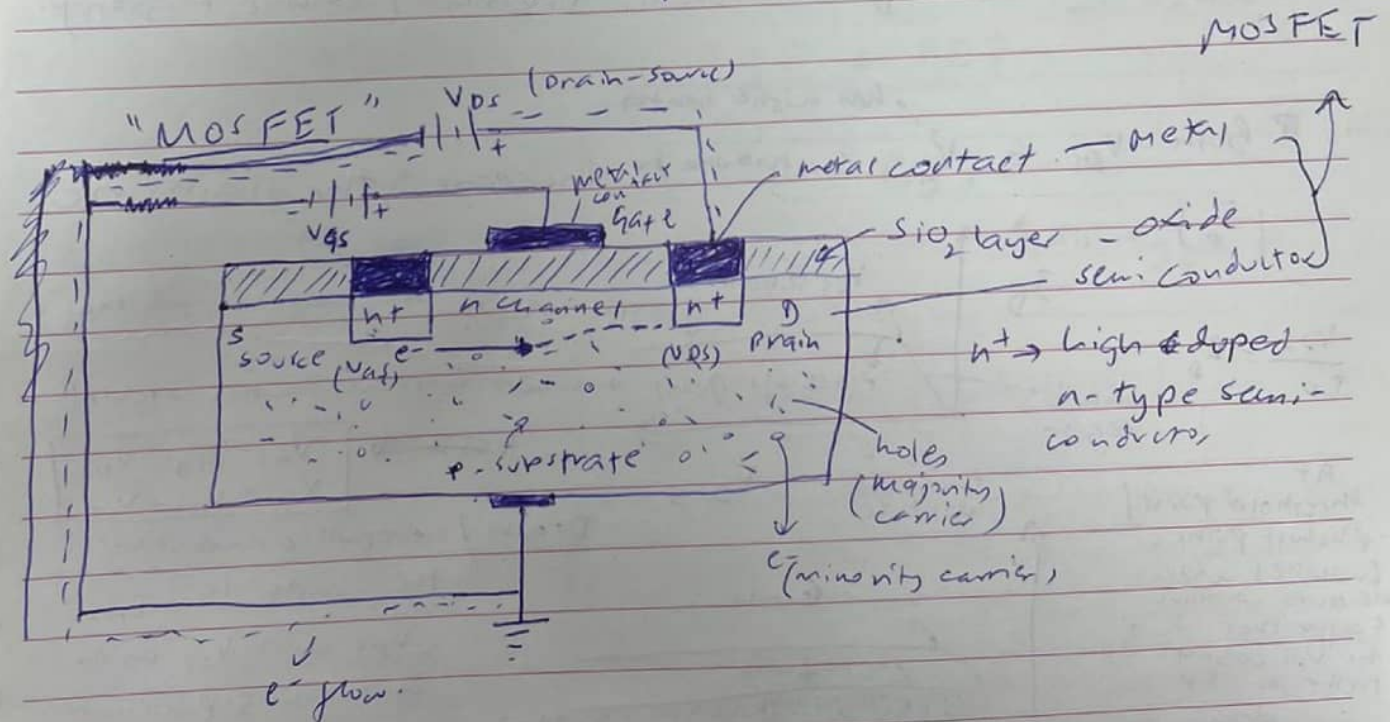
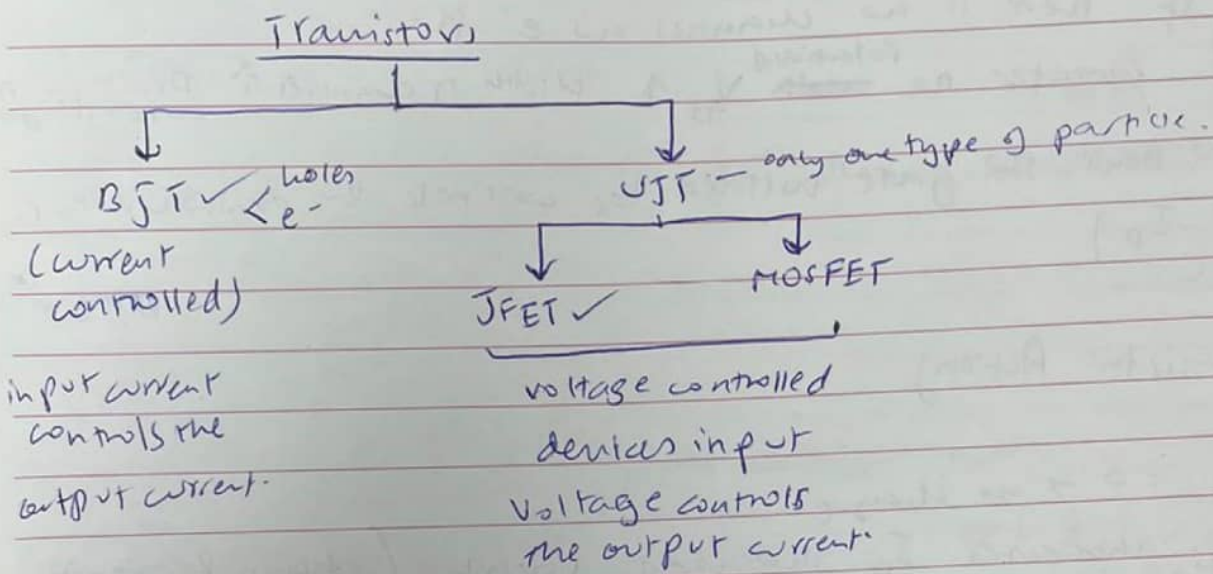
- Collector base junction is reverse

- 4 possibilities are; 1) E-B-f.b

- 2) E.B - ~~Ab~~
- 3) B.C - fb
- 4) B.C - rb

Increasing the Drain voltage will increase the current, and also make the channel narrower. At a certain point, current becomes constant. It is called Pinch-off voltage.

→ Metal oxide Silicon Field Electric Transistor (MOSFET)



V_{DS} = External battery b/w Drain + Source
 V_{GS} = " " " Gate + Source

day / date:



V_{GS} attracts the e^- in p-substrate region, due to which e^- collect near the SiO_2 layer, forming a channel. This is N-channel
 ↳ as it is channel of e^-

When V_{DS} attracts the e^- and forms a path a circuit from the close circuit. from n^+ (source) to n^+ (drain).
 ↳ e^- are majority

* If there is no channel no e^- flow.

Greater the ^{Potential} V_{GS} ↑ Width of channel ↑ Drain current I_D ↑

∴ Here the gate voltage V_{GS} controls the drain current (I_D)

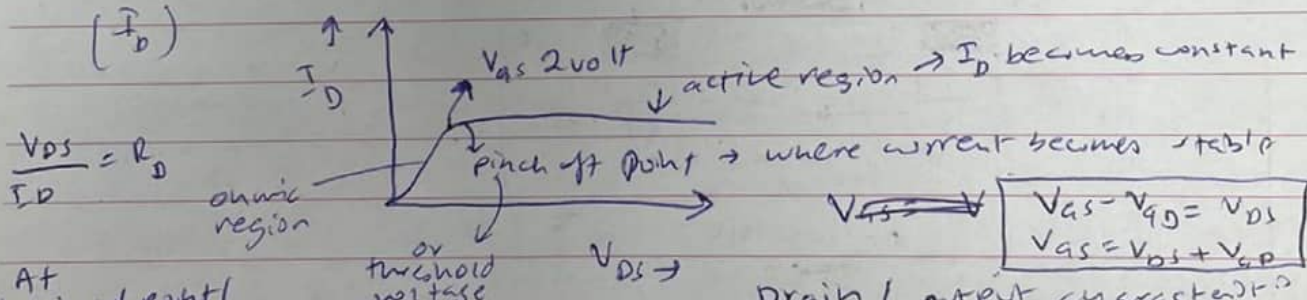
(Transistor Action)

* $V_{GS} = 0 \rightarrow$ no channel

V_{GS} increases I_D increases linearly (ohmic region)

* Both V_{DS} & V_{GS} contribute to increase of the drain current

(I_D)



At threshold point / pinch off point, channel width becomes constant & after that inc. V_{DS} doesn't further inc. I_D

ohmic region

as acting as a resistor $V \propto I$

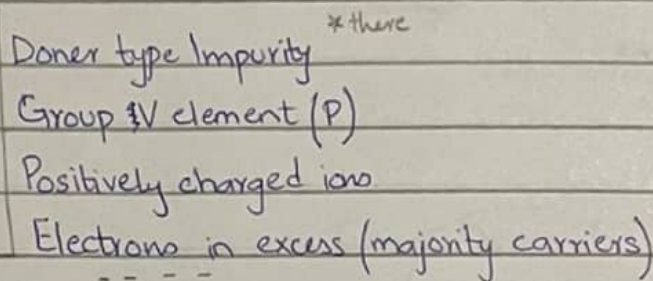
Drain / output characteristic graph b/w I_D & V_{DS} with V_{GS} as the controlling parameter.

V_{DS} current constant



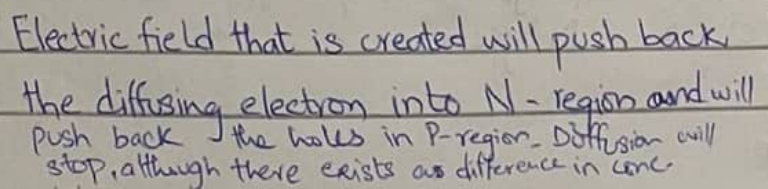
:- नक्षत्र 14-9

Unbiased p-n junction:- (no ^{external} voltage is applied)


$$\begin{array}{ccccc} + & + & + & + & + \\ + & + & + & + & + \\ \hline + & + & + & + & + \end{array}$$

→ electrons combines with a hole and both are annihilated

Positive " " N-type " " " (" " " " " electrons)



Channel width is enhanced when V_{GS} is increased. Hence it is called Enhancement Mode MOSFET.

Depletion Mode MOSFET. $\rightarrow$ when channel width is decreased when V_{GS} is increased.

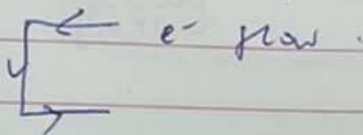
P-channel MOSFET
enhancement mode

$n^+ \rightarrow p^+$

p-substrate $\rightarrow$ n-substrate

$|+ \rightarrow |-$

$|- \rightarrow |+$



channel $\rightarrow$ majority carrier holes.

CMOS
 $\downarrow$
complementary.

$\rightarrow$ I.C $\rightarrow$ $\rightarrow$ circuit in which all transistors are on one chip. multiple

COMPARISON B/W BJT & FET

FET	BJT
1) involves only one type of charges.	1) involves two types of charges
2) voltage controlled device	2) current controlled device.
3) simple to fabricate consumes less space $\downarrow$ hence MOSFET stage in Integrated Circuit. $\rightarrow$ longer battery life	3) difficult to fabricate consumes more space in I.C $\rightarrow$ circuit in which all transistors are present on one chip.
4) consumes less power (10 times less than BJT) Cost is less	4) consumes more power
	5) Cost is less

⇒ Intrinsic semiconductor :- $J_T = (n \cdot q \cdot \mu_n \cdot E) + (p \cdot q \cdot \mu_p \cdot E)$

$n \cdot p = n_i^2 \Rightarrow$ Law of mass action (always true, even after adding impurity)

$= n_i \cdot q \cdot \mu_n \cdot E + n_i \cdot q \cdot \mu_p \cdot E$

$= n_i \cdot q \cdot (\mu_n + \mu_p) E$

Movement of subatomic particle :-

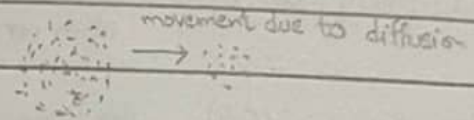
Drift

(for electrical charges
Like e^- , holes)
is dependent upon
applied voltage/electrical
field

$$J = \sigma E$$

Diffusion

(for non-electrical particles)
due to difference in concentration



Current Density $J \propto \frac{dN}{dx} \rightarrow$ conc. gradient

In a Extrinsic semiconductor (N-type) :-

→ e^- in conduction band

→ holes in valence "

current due to electrons = drift current + diffusion current

$$= J_{n1} + J_{n2}$$

$$= n \cdot q \cdot \mu_n \cdot E + q \cdot D_n \frac{dn}{dx}$$

$$J = qD \cdot \frac{dN}{dx}$$

where D is the diffusion coefficient

conc. is ~~not~~ uniform

$$J = 0$$

current due to holes = $J_{p1} + J_{p2}$

$$= n \cdot q \cdot \mu_p \cdot E + q \cdot D_p \frac{dp}{dx}$$

$$J = q \cdot D \frac{dN}{dx}$$

$$J_n = q D_n \cdot \frac{dn}{dx} ; \text{electrons}$$

$$J_p = q D_p \cdot \frac{dp}{dx} ; \text{holes}$$

Total current density = $J_n + J_p$ Amps/cm²

P-N types:-

N-type semiconductor:-

- Impurity: Group V elements (P) N_D / cm^3
- majority carries = n (conduction band)
- minority carries = p (valence band)

$$n \approx N_D^{\text{Donor}} / \text{cm}^3$$

$$p \approx \frac{n_i^2}{N_D} / \text{cm}^3$$

P-type semiconductor:-

- Impurity: Group IV element (A) N_A / cm^3
- majority carries = p (valence band)
- minority carries = n (conduction band)

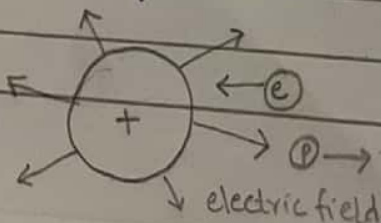
$$p = N_A^{\text{acceptor}} / \text{cm}^3$$

$$n = \frac{n_i^2}{N_A} / \text{cm}^3$$

Movement of subatomic particles:-

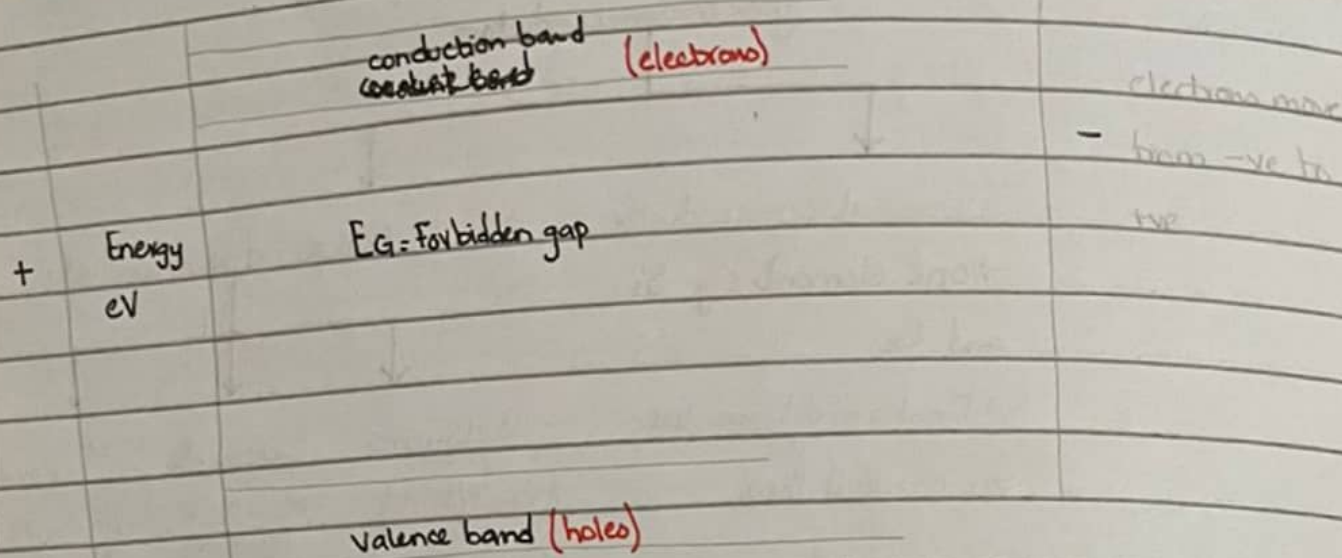
Drift

- Movement of particle as a result of electric field
- electrical phenomenon



Diffusion

- Movement of particles due to difference in concentration
- non-electrical phenomenon
- Smoke, dust, fluid } Brownian movement
- * - movement from higher concentration to lower concentration. Movement will stop when " " is equalised



$$I_T = I_p + I_e$$

↓
holes

The mobility in the conduction band is much higher than mobility in valence band (free electron)

↓

The ability of electrons to move

μ_e = mobility of electrons

μ_p = " " holes

(stop to stop or deficiency to fill)

Recombination:- Electrons fall back into valence band when temp. decreases and form covalent bonds again.

Relation b/w conductivity and mobility:-

Conductivity = ability to conduct electricity

Mobility = " of electrons to move

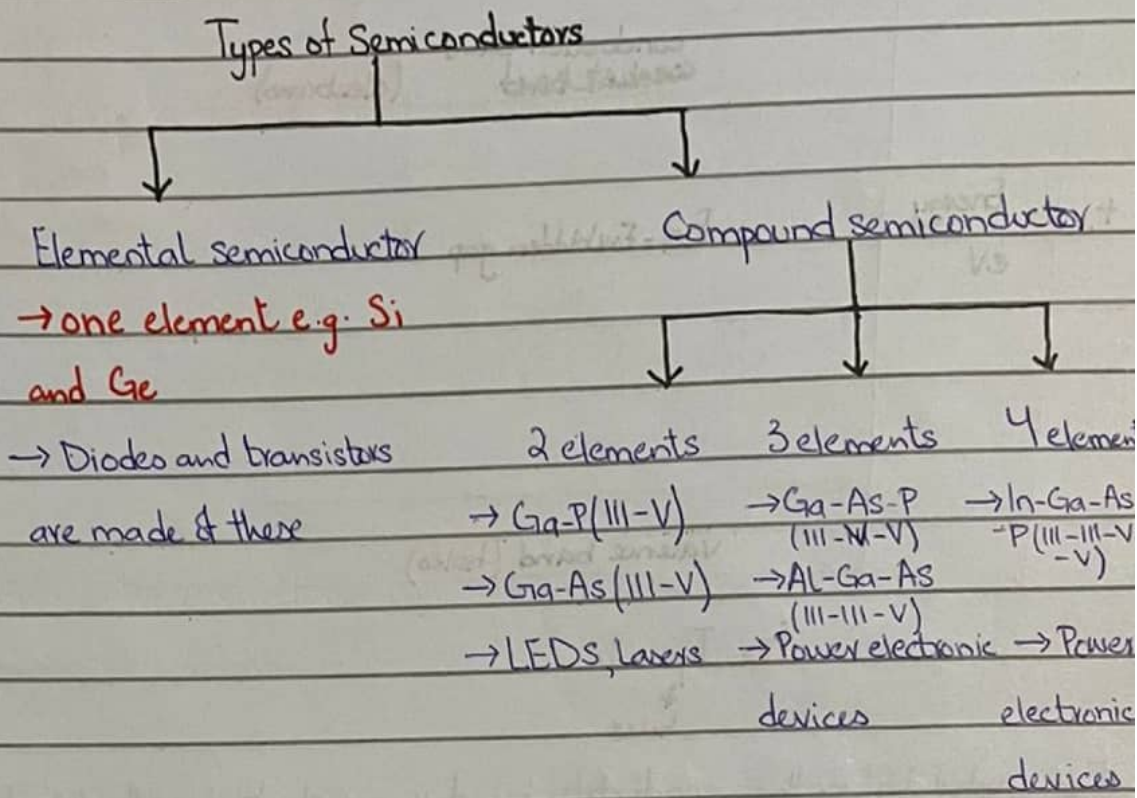
$$\sigma = n * q * \mu$$

conductivity = no. of electrons x charge of e^- x mobility of e^-

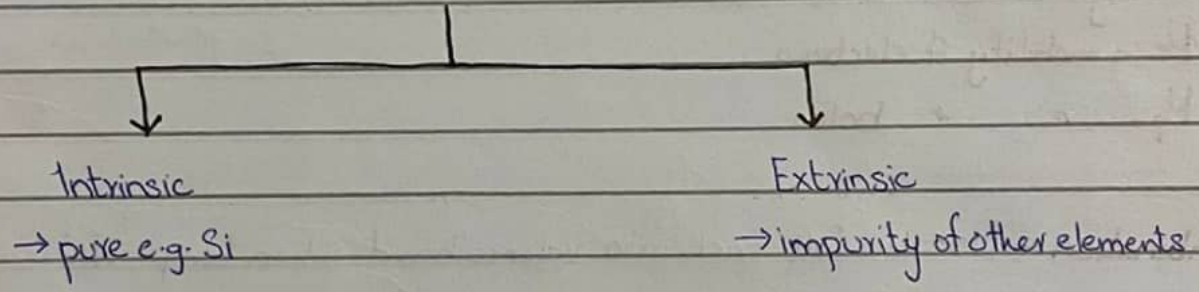
$$\frac{1}{\text{ohm}\cdot\text{cm}} = \frac{1/\text{cm}^3}{(\text{concentration of } e^-)} \times \text{coulomb} \times \frac{\text{cm}^2}{\text{volt}\cdot\text{sec}}$$

$$\frac{1}{\text{cm}^3} \times \text{A}\cdot\text{s} \times \frac{\text{cm}^2}{\text{ohm}\cdot\text{cm}} \sim \text{A}\cdot\text{s}$$

Conductors

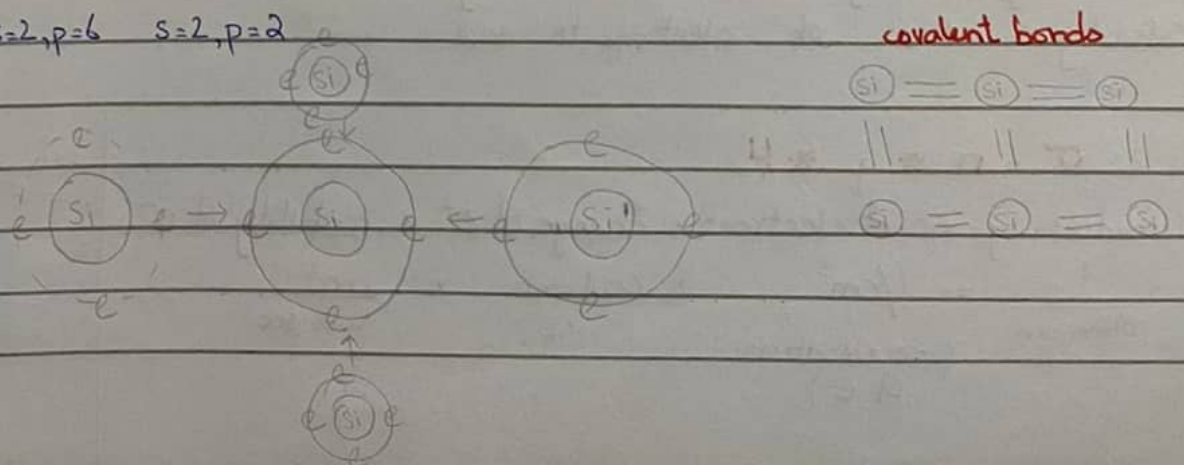


Types of semi conductors with reference to pure or impurity :-



Si¹⁴ e = 14e (valence shell) imp. in terms of conductivity

	K	L	M	N
shells	2	8	4	
orbital	s=2	s=2, p=6	s=2, p=2	



Insulator :-

Neon; ^{10}Ne $e = 10$

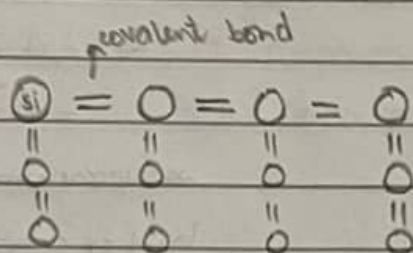
K	L	M	N
2	8		
$s=2$	$s=2, p=6$		
stable	stable		

Inert; apply current but nothing happens since no electrons available

Semiconductor :-

Silicon; ^{14}Si $e = 14$

K	L	M	N
2	8	4	
$s=2$	$s=2, p=6$	$s=2, p=2$	
		incomplete orbital	



The silicon atoms achieve stable structure by sharing electrons from their neighbour atoms to form covalent bonds. The outermost electron cannot readily participate in the conduction. Hence conductivity of semiconductor is very small in comparison

⇒ When heat is applied to semiconductor, the heat energy is converted to kinetic energy of electrons and covalent bonds start breaking. Electrons become free and are available for conduction. Hence conductivity of semiconductor increases with temperature

to conductor at room temperature $\approx 25^\circ\text{C}$ to 30°C

⇒ current can only flow if free electrons are available

Silicon :-

Atomic No. = 14

	K	L	M	N
Shells	2	8	4	
Orbital	(s=2)	(s=2, p=6)	(s=2, p=2)	

[Explanation same as germanium]

C Copper :-

Atomic No. 29, $p=29$, $e=29$

H	K	L	M	N	
	2	8	18	31	shells

(s=2) (s=2, p=6) (s=2, p=6, d¹⁰) (s=1, p=1) orbitals

D stable stable stable unstable → because of this copper is good conductor as it is incomplete orbital in outer shell

P Argon (Ar) :-

Atomic No. = 18

H	K	L	M	N
	2 stable	8 stable	8 unstable	

(s=2) stable (s=2, p=6) stable (s=2, p=6) stable

(According to Rutherford it is unstable orbital (According to Plank it is stable)

Hence it is inert and doesnot reactive with others. This is insulator that doesnot allow flow of current

Germanium (Ge) :-

Atomic No. 32

H	K	L	M	N
shells	2	8	18	4

orbitals (s=2) (s=2, p=6) (s=2, p=6, d=6) (s=2, p=2)

(According to rutherford, this is unstable)

↑ This can result in conduction (small conduction)

↓ cause of bond ⇒ covalent
↓ doesnot allow good conduction
hence semi conductor

Crystal

Hole

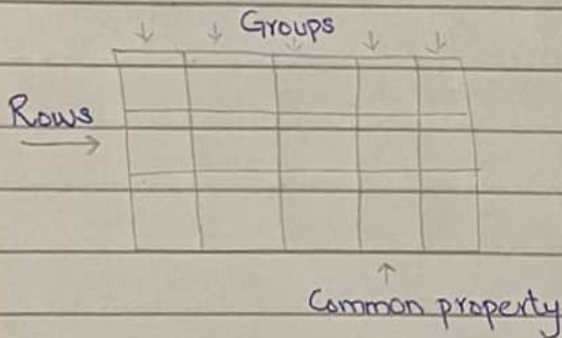
Doping

P-N junction

Bernex Metial

Periodic table of Elements :- (Refer google)

Group 3A, 5A, 6A, 4A are semiconductor



Flow of Electricity / Electrical conduction :-

1) Insulator; donot allow current to flow e.g. carbon, glass, inert gas (nitrogen) (helium)

~~2) Class~~

a) Conductor; freely allow current to flow e.g. all metals

How AC devices are different from DC devices?

Electricity (AC)

- High voltage $> (220V)$
- Current greater than 1 Amp
- power greater than 200W
- e.g. generator, transformer, big motors

Both obeys
ohm's Law

Electronics (DC)

- Low voltage $< (30V)$
- current lesser than 1 Amp
- power lesser than 10W
- e.g. mobile phones, TVs, videos, network devices, routers

⇒ higher voltages, more unsafe
↓
our blood is electrolysed and causes clotting.

Terms :- (refer google)

Atom; An atom is a particle of matter that uniquely defines a chemical element.

It consists of a central nucleus^(+ve) that is surrounded by 1 or more -vely charged electrons.

Nucleus has 1 or more relatively heavy particles known as protons and neutrons.

Photons; A photon is a tiny particle that comprises waves of electromagnetic radiation. Photons have no charge, no resting mass and travel at the speed of light.

Electrons; An electron is a -vely charged ~~particle~~ subatomic particle that can be either bound to an atom or free. Together, protons and electrons form the atom's nucleus.

Shells; According to Bohr, electrons revolve around the nucleus in a specific circular path known as orbit or called a shell. They have stationary energy levels, the energy of each shell is constant.

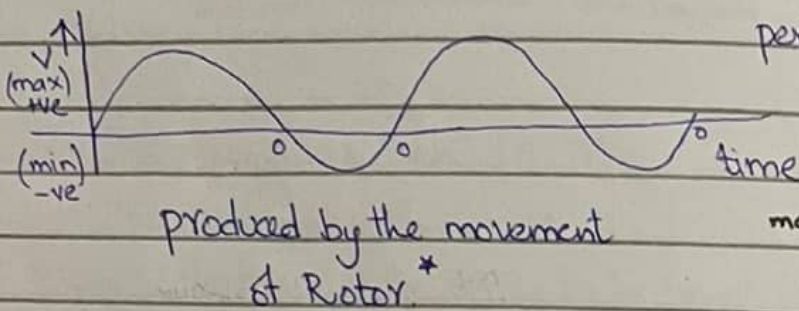
Valence Electrons; Valence electrons are the electrons in the outermost shell, or energy level, of an atom.

* moves from turbine in dams.

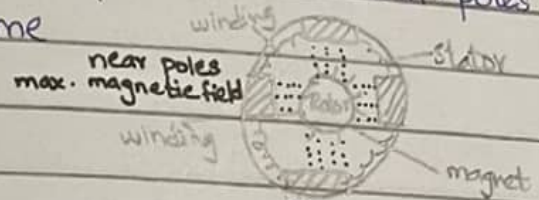
AC and DC voltages and currents

Alternate current and voltage :-

- Magnitude of voltage with time (according to sine wave)
- Direction of voltage changes for +ve and -ve and vice versa



periodic sine wave
AC
(Generator have diff. poles)



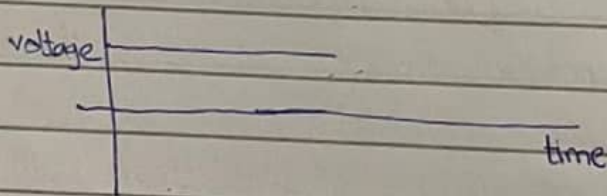
* change is what gives AC bigger hand than DC

rotor when moves in the magnetic field
voltage/EMF is produce (Faraday's Law) (Electro magnetic induction)
$$EMF = -N \frac{d\phi}{dt}$$

no. of turns $\rightarrow$ value of change of flux $\rightarrow$

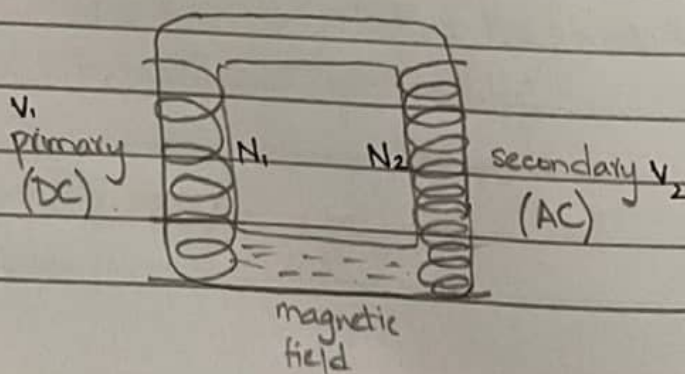
Direct current and voltage :-

- No change with time



* [induced emf generated in AC ($\frac{d\phi}{dt}$) but not in DC ($\frac{d\phi}{dt} = 0$)]

[if there is no $\frac{d\phi}{dt}$ there is no transformer action that is based on electromagnetic induce hence no emf in DC]



$V_2 \propto \frac{N_2}{N_1}$
 V_1 hence $V_2 \uparrow$ increase N_1 increased N_2
(step up and down is done by N_2 and N_1)

Q = Find the nearest star in light years to the distance of the Earth.

$$\text{Light year} = 9.461 \times 10^{15} \text{ m}$$

Units of measurement :-

Voltage ; capacity to do ^{energy} work / unit of energy (more distance more energy / potential)

Current ; ^{(coulomb/second)* = amps} energy in moving the charge (-ve to -ve [repelling]) (the amount of work done to bring one coulomb to another 1c is 1 volt) [flow of charge / rate of flow of charge*]

Power ; energy divided by time = Joules/sec = watt

Energy ; Joules

Resistance ; opposition to the flow of current / measured in ohm $\Rightarrow \frac{\text{Volts}}{\text{amps}}$

smaller resistance move current $V \propto I(\text{current})$ [V $\Rightarrow$ cause]
vise versa $\rightarrow$ current won't pass if no ^{*}resistance $\rightarrow$ [I $\Rightarrow$ effect]
infinite resistance

~~3~~ Fundamental energies :-

1) Gravitational force (due to gravity)

2) Electrical force (due to charge)

3) Magnetic force (due to magnetism)

E.M waves from electrical and magnetic force.

eVolts = Joules / charge of electron

1) Convert the following in engineering notation :-

i) 1.500

$1.5 \times 10^3 \Rightarrow 1.5 \times E3$

ii) 63,200,000

$63.2 \times 10^6 \Rightarrow 63.2 E6$

iii) 0.0234

$23.4 \times 10^{-3} \Rightarrow 23.4 E-3$

iv) 0.000059

$0.059 \times 10^{-3} \Rightarrow 59 E-6$

2) Convert the following into decimal notation :-

i) $1.2 E3$

1200

ii) $547 E6$

547000000

iii) $2 E^{-3}$

0.002

(iv) $27 E-9$

0.000000027

v) $4.39 E7$

43900000

3) Use appropriate Prefixes :-

i) $4 E6$ volts

4 Mega Volts

4×10^6 volts

ii) $5.1 E3$ feet

5.1×10^3 f

5.1 kft

iii) $3.3 E-6$ grams

3.3 micrograms

4) Perform the following operations :-

i) $5.2 E6 + 1.7 E6 \Rightarrow 6.9 E6$

ii) $12 E3 - 900 \Rightarrow 12 E3 - 0.9 E3 \Rightarrow 11.1 E3$

iii) $1.7 E3 \times 2 E6 \Rightarrow 3.4 E9$

iv) $48 E3 / 4 E6 \Rightarrow 12 E-3$

v) $20 / 4 E3 \Rightarrow 5 E-3$

vi) $10 M \times 2 K \Rightarrow 20 E9$

vii) $8 n / 2 M \Rightarrow 4 E-6$

Some physical Constant :-

$$\text{Speed of light } (c) = 2.998 \times 10^8 \text{ m/s} = 3 \times 10^8 \text{ m/s} = 0.3 \times 10^9 \text{ m/s} = 0.329 = 0.3 \text{ Giga} \frac{\text{m}}{\text{s}}$$

$$\text{Charge of the } e = -1.6 \times 10^{-19} \text{ Columb}$$

$$\text{charge of proton} = +1.6 \times 10^{-19} \text{ columb}$$

$$\text{Mass of electron} = 9.107 \times 10^{-31} \text{ Kg}$$

$$\text{Mass of proton} = 1.672 \times 10^{-27} \text{ Kg}$$

$$\text{charge to mass ratio } (e/m) = 1.7 \times 10^{11} \text{ C/Kg}$$

$$\text{Boltzmann's constant } (k) = 1.38 \times 10^{-23} \text{ J/K} \rightarrow \text{Kelvin}$$

$$\text{Plank constant } (h) = 6.547 \times 10^{-34} \text{ J.s}$$

$$\text{Permittivity of free space } \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$\text{Permeability of free space } \mu_0 = 1.257 \times 10^{-6} \text{ H/m}$$

Henry (conductor)

$$\text{Impeadance of free space} = Z_0 = 376.7 \text{ ohm}$$

$$\text{Thermal voltage } (kT) = 0.0259 \text{ eV} \rightarrow (\text{energy expended in moving an electron})$$

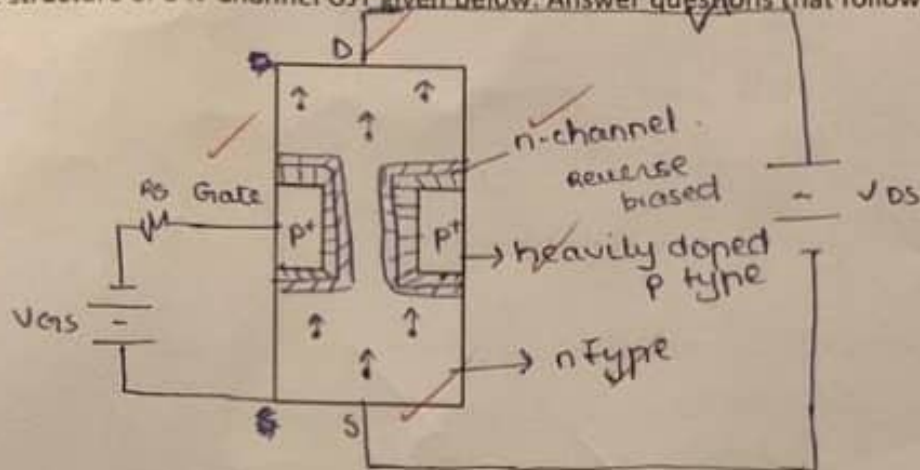
boltzmann constant

$$\text{Power } (P) = \text{volts } (V) \times \text{Amps } (A)$$

$$\begin{aligned} \text{Energy } (J) &= \text{Power } (P) \times \text{time } (t) \quad [\text{rate of power}] \quad (V \times A \times t) \\ &= \text{volts} \times \frac{(\text{columb})}{\text{time}} \times \text{time} \end{aligned}$$

$$= \text{volts} \times \text{columb} \rightarrow \text{volts} = \text{Joules/...}$$

Consider the structure of a N-Channel UJT given below. Answer questions that follow.

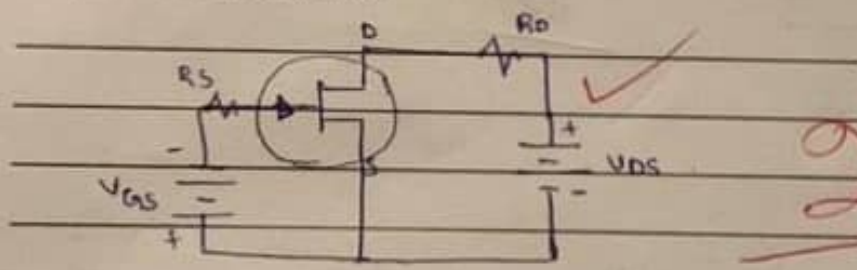


Mark the different regions as N type or P type. Mark the regions source, gate and drain.



2

Draw the circuit symbol of the above transistor and connect power supply to input and output circuits.



2

How does the UJT compare with BJT in terms of (a) fabrication area (b) power consumption.

- (a) BJT has low fabrication area and UJT has higher fabrication area.
- (b) BJT has high power consumption. UJT has low power consumption

2

The waveform in Fig 1 below is a sinusoidal waveform with maximum value of 20 Volts and frequency of 50 Hz. The waveform in Fig 2 is a rectified waveform with maximum value of 20 volts. Answer the questions given below.

Fig 1

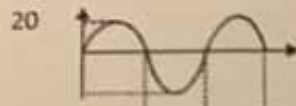
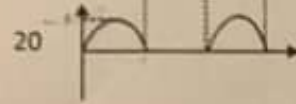


Fig 2



Find the Average and RMS value of waveform in Fig 1.

$$Avg = 0$$

$$V_{rms} = 20/\sqrt{2} = 14.1 \text{ volts}$$

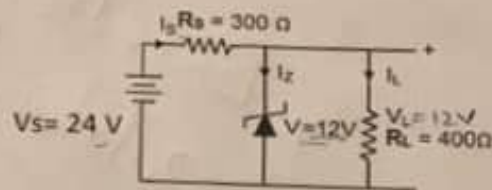
Find the Average value of waveform in Fig. 2

$$V_{avg} = 20/3.142 = 6.37 \text{ volts}$$

Find the Time Period of waveform in Fig 2.

$$T = 1/50 = 0.02 \text{ s}$$

With ref to Zener Diode Circuit answer the questions that follow:



Calculate the value of I_L

$$V_L = 12V$$

$$I_L = 12/400 = 0.03 \text{ amps}$$

Calculate the value of I_Z

$$V_Z = 12V$$

$$V_{RS} = 24 - 12 = 12V$$

$$I_Z = 12/300$$

$$I_S = 12/300 = 0.04 \text{ amp}$$

$$I_Z = 0.04 - 0.03 = 0.01 \text{ amp}$$

Can the above Zener Diode regulate voltage of 10 V across the load.

Explain your answer.

No. A Zener diode cannot be made adjustable. It is specific to its voltage. To regulate voltage of 10V a different Zener diode can be used.

1/✓ In a P-type semiconductor what is the meaning of p and N_A .

p is referred to the majority carrier concentration, hole concentration

N_A is the concentration of acceptor impurity.

$$p \approx N_A$$

1/✓ Why is a P-Type Semiconductor electrically neutral.

It's electrically neutral because although due to borrowing of e^- there is a negative charge on the acceptor impurity atom there is also a positive charge on the silicon atom which donated the e^- . There is only a redistribution of charge, there is no net charge.

1/✓ Why is mobility of holes less than mobility of electrons?

$\mu_n > \mu_p$ because electrons are present in the conduction band where they are free and holes are present in valence band where they move 1 step at a time.

1/✓ At what stage does Diffusion take place in manufacture of P-N junction?

Diffusion takes place in an unbiased P-N junction when no potential difference is applied. This right after the manufacturing process.

1/✓ What is the function of Junction Potential?

Junction Potential is an area where E is generated & it pushes the diffusing e^- back into the N region & pushes the diffusing holes back into the P region.

$$p = 2.25 \times 10^{15}$$

$$n_i = 1.5 \times 10^{10}$$

1/✓ In a p-type Silicon sample, the hole concentration is $2.25 \times 10^{15} / \text{cm}^3$. The intrinsic carrier concentration is $1.5 \times 10^{10} / \text{cm}^3$. Find the concentration of electrons.

$$p \approx N_A$$

$$n_n = \frac{(n_i)^2}{N_A} = \frac{(1.5 \times 10^{10})^2}{(2.25 \times 10^{15})} = \frac{(2.25 \times 10^{20})}{(2.25 \times 10^{15})} = 1 \times 10^5 \checkmark$$

$$n_n = 1 \times 10^5 / \text{cm}^3$$

1/✓ Given that $v = \mu E$ where v is the drift velocity, μ is the mobility in $\text{cm}^2 / \text{volt} \cdot \text{sec}$ and E is the Electric Intensity in volt/cm . Work out units of v .

$$v = \mu E$$

$$= \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \times \frac{\text{V}}{\text{cm}}$$

$$v = \text{cm} \cdot \text{s}^{-1} = \text{cm/s} \checkmark$$

1/✓ Express Current Density J in terms of Coulombs.

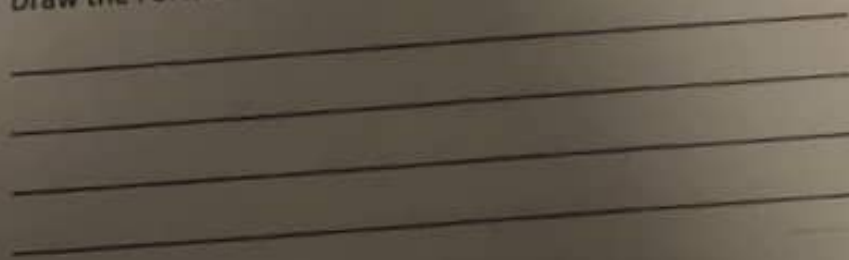
$$J = n q \mu E$$

$$= n q \checkmark$$

$$= \frac{\text{C}}{\text{cm}^2 \cdot \text{s}} \times \frac{\text{C}}{\text{s}} \times \frac{\text{cm}}{\text{s}} \checkmark$$

$$= \frac{\text{C}}{\text{cm}^2 \cdot \text{s}} \checkmark$$

0.5/ Draw the Forward characteristics of a Silicon P-N Junction showing Cut-in voltage.



value of cut-in voltage?

cut in voltage

1/✓ Why is Cut-in Voltage equal to Barrier Potential (V_{bi}) in a PN junction?

Cut in voltage is when external voltage has completely override the effect of V_{bi} . Thus after Cut in voltage graph increase exponentially as V_{bi} has no effect.

